Supply Chain Analytics: A Data-Driven Approach

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Contents

Chapter 1

Demand Estimation and Forecasting

1.1 Introduction

Demand forecasting is an important tool to make decisions such as financial projections, capacity planning and allocation of resources for a productive activity. The demand for any product, service or good in the future is a random variable. That is, it is unknowable before it is realized. The demand that already occurred is called *realized demand* or can also be called historical demand.

In fact, only the past is really known, everything happening in the future has to be guessed. In these lectures notes we deal with the most common practices in *guessing* the future quantities of the demand. Mathematically, it is useful to model the demand as a random variable. Random variables are a mathematical construct to capture events and their randomness. We will cover the basics building up the necessary knowledge to manipulate and eventually model and estimate demand in the future.

1.2 The Demand as a Random Variable

Suppose we want to model the demand quantity once in the future (say, next week or next month or next year). We denote this random quantity by the capital letter D. This random variable D is in fact a function that takes as input a random mechanism ω and outputs an event, in our case a number for the demand[1]. For example, the demand can be a number like $D = 14$ with probability p_{14}, or it could be $D = 5,000$ with a probability $p_{5,000}$.

As normally the demand is a *count* (of products, goods or services) we will assume the demand cannot take negative values. Then, we can describe the probability of any outcome

[1] The space of random mechanism is normally called sample space, and is denoted as the set Ω. Then, formally the random variable is a function mapping $D : \Omega \to E$ to a set of possible events E.

of the demand by a *probability mass function* or pmf as $p_i = \mathsf{P}(D = i)$ for all possible values $i = 0, 1, \ldots$ For the pmf to be correctly defined two things must happen: First, each outcome must occur with a probability between 0 and 1, that is $0 \leq p_i \leq 1$ and adding over all outcomes should add up to 1, that is, $\sum_{i=0}^{\infty} p_i = 1$.

With this framework, we can already solve interesting questions: For example, if D is the quantity of semiconductor chips sold in the next quarter, we can ask *what is the probability that the demand is at least 10 million chips?*. To answer this, the event $\{D > 10^7\}$ is the same as the events $\{D = 10^7 + 1, D = 10^7 + 2, ...\}$, then we can simply do:

$$\mathsf{P}(D > 10^7) = \mathsf{P}(D = 10^7 + 1, D = 10^7 + 2, \ldots) = \sum_{i=10^7+1}^{\infty} p_i, \tag{1.1}$$

Throughout the topic it is often useful to get used to manipulate event sets. For example, in the previous example we exploited the fact that the event $\{D > 10^7\}$ can be written as the union that the demand takes exactly each value greater than 10^7, noting that each of these events is disjoint (it can be a value or another but not two at the same time). We could have also taken an alternative approach by calculating the probability of the *complement* event, that is, that the demand is less or equal to 10^7 and then, given that the probability of all outcomes is 1, subtract $P(D \leq 10^7)$. In other words:

$$\mathsf{P}(D > 10^7) = 1 - \mathsf{P}(D \leq 10^7). \tag{1.2}$$

For a given number i, the probability $F_D(i) = \mathsf{P}(D \leq i)$ is known as the *cumulative mass function* or cmf. The usefulness of the cmf, is related again to event calculus, for example asking the probability that the demand is between 1 and 10 million units. That is, $\mathsf{P}(10^6 < D \leq 10^7)$ can be expressed in terms of the cmf as $F_D(10^7) - F_D(10^6)$.

Another very important quantity is known as the mean, or average. This comes from averaging all possible outcomes for the demand weighted by their probabilities. This quantity is denoted as $\mu_D = \mathsf{E}(D) = \sum_{i=0}^{\infty} i p_i = 0p_0 + 1p_1 + 2p_2 + \ldots$ [2] Averages are important for demand estimation, because they provide a scenario that is centered around the more likely outcomes for the demand. For example, when making a resource allocation decision, such as building a factory or a data-center, the capacity should comfortably handle the average demand for the facility. But then, the next immediate question is how much slack capacity should the factory have?

The variance of a random variable is a measure of how much the random variable *deviates* from its mean μ_D. The variance is defined as $\sigma_D^2 = \mathsf{Var}(D) = \mathsf{E}[(D - \mu_D)^2]$, the E operator (or function) denotes the operation of multiplying each possibly outcome by its probability. So, $\mathsf{E}[(D - \mu_D)^2]$ can be computed as $\sum_{i=0}^{\infty} (i - \mu_D)^2 p_i$. The square root of the variance is known as the *standard deviation* as is denoted by the symbol σ_D.

[2]Related to event calculus, the mean μ_D is also equal to $\sum_{i=0}^{\infty} \bar{F}_D(i)$, where $\bar{F}_D(i) = 1 - F_D(i)$. Why is this the case?

1.2.1 Useful Probability Calculations

So far, we have reviewed basic concepts in probability that can answer some questions about the random outcome of the demand. We can go one step further and with some knowledge of the distribution, answer other questions: For example, say that you only know the mean μ_D of the demand, and they ask you, if the average demand is 1 million units, what's the probability that the demand is less than 5 million?

A useful relation known as *Markov's Inequality* states that $\mathsf{P}(D \geq a) \leq \mu_D/a$, replacing $a = 5 \times 10^6$ tells us that the probability that the demand exceeds 5 million units is at most $\mu_D/(5 \times 10^6)$. If the average demand μ_D is 1 million, the probability that it is at least 5 million is at most $\mu_D/a = 10^6/(5 \times 10^6) = 1/5 = 20\%$. Therefore, the probability that the demand is less than 5 million is at least 80% (could be more). That being said, this estimate is rather loose as the inequality is *at most* the value. In the previous example, it could be the case that the probability that the production exceeds 5 million is exactly 0.

Another useful inequality is known as *Chebyschev's Inequality* stating that $\mathsf{P}(|\mu_D - D| \geq k\sigma_D) \leq 1/k^2$. Going back to our previous example, suppose that additionally to knowing that the average demand μ_D is 1 million, we also know that its standard deviation σ_D is 100,000 units. Applying this inequality, the probability that the demand is below 800,000 or above 1,200,000 is at most 25% (using the inequality with $k = 2$). Going back to the previous question about the probability that the demand exceeds 5 million is at most $1/40^2 = 0.06\%$. Even more can be said, for example the probability that the demand exceeds 2 million is less than $1/100 = 1\%$. The moral of this, is that with just the mean μ_D and the standard deviation of the demand σ_D there is already a lot of useful information that can be inferred about the demand without knowing the full probability distribution of all its outcomes.

1.2.2 Side Information

The demand is clearly influenced by other variables (random or deterministic): the economic cycle, interest rates, competitors prices, the demand of other products, weather and so on. Having a framework to understand how a variable affects the demand is crucial to better estimate and anticipate the demand.

Suppose you are in charge of the production of the Airpods wireless earphones. Intuitively, the demand of Airpods is *positively* related to the demand of other products in the iOS ecosystem (such as laptops, iPhones, etc) and *negatively* to competitor products, such as Android headphones. Moreover, the demand for Airpods should not be affected by the demand of apples or oranges.

Ideally, it would be great to have a mathematical function g, that given an outcome of a related random variable $X = x$ gives us the best estimate of the random demand D. You can imagine X as the demand for iPhones in the AirPods example (as a significant portion of AirPods users own iPhones). Then, we would like to find the function g that in average gives the closest prediction of the demand D. One such formulation is

$$\min_g \mathsf{E}\left[(g(X)-D)^2\right], \tag{1.3}$$

This way of thinking is a strategy that we will repeatedly use in the exposition of the topic. That is, to ask what's the best possible data-driven way to make guesses about the random nature of the demand. Assuming X can take $x_1, x_2, \ldots$ different values. The expected value $\mathsf{E}\left[(g(X)-D)^2\right]$, taking $p_{ix} = \mathsf{P}(D=i, X=x)$ can be written as:

$$\mathsf{E}\left[(g(X)-D)^2\right] = \sum_{i=0}^{\infty} \sum_{x=x_1,\ldots} (g(x)-i)^2 p_{ix}, \tag{1.4}$$

The strategy to solve a seemingly complex problem like this is to break down a complex problem into smaller and easier problems to solve (a well known paradigm in Computer Science known as divide-and-conquer). Then, the first observation is that the solution only depends on the values of the random variable X. That is, for a given $X=x$, the function $g(x)$ is a single value guessing the most likely outcome of the demand D. Taking only a single value of $X=x$ gives the expression $\sum_{i=0}^{\infty}(g(x)-i)^2 p_{ix}$. This expression has a single unknown $g(x)$, which is a convex function[3]. Taking derivatives and equating to 0 leads to the following expression for $g(x)$:

$$g(x) = \sum_{i=0}^{\infty} i\frac{p_{ix}}{p_x} = \mathsf{E}(D|X=x),$$

This expression intuitively says, that the best predictor for the demand D when the side variable X is equal to x is the expected value of the demand given that $X=x$. This is the so-called *conditional expectation*, which can be seen as simply taking the well known expected value with new probabilities p_{ix}/p_x representing the likelihood of each outcome of the demand given a certain outcome for the side variable X. The denominator p_x scales the new probabilities p_{ix} of the demand so that they add up to 1, because instead of considering all outcomes of X, only the case where $X=x$ is considered.

Example 1.1 (Hydroelectric Plant). Suppose you are managing a hydroelectric plant supplying electric energy to a major city. X represents whether or not it is an extremely hot day ($X=1$ means it is extremely hot, and $X=0$ means it is not), extremely hot days occur rarely, say $\mathsf{P}(X=1)=p_h=5\%$. But when they occur, the demand for electricity is very high (due to the usage of Air Conditioner and other appliances). The conditional distribution of $(D|X)$ is given by $\mathsf{P}(D=5\text{Mw}|X=0) = \mathsf{P}(D=15\text{Mw}|X=1) = 1$. Given this distribution, what is

[3]A sum of convex functions is convex.

the average demand μ_D of the power plant? The average demand of the power plant is:

$$\mu_D = \mathsf{E}[D] = \mathsf{E}[\mathsf{E}[D|X]] = 5\text{Mw}(1 - p_h) + 15\text{Mw}p_h = 5.5\text{Mw}. \quad (1.5)$$

This relationship clearly shows one of the main ideas of this course: While the demand in average is relatively stable at a level of 5.5Mw most days, with a small probability it can jump to almost a 300% increase if it's a very hot day. If the power plant does not have the capacity to service the demand, it can be a catastrophic event for the city, since many crucial services including hospitals and other vital supply chains rely on electricity. Hedging against rare events is the whole point of a resilient supply chain. See Figure 1.1.

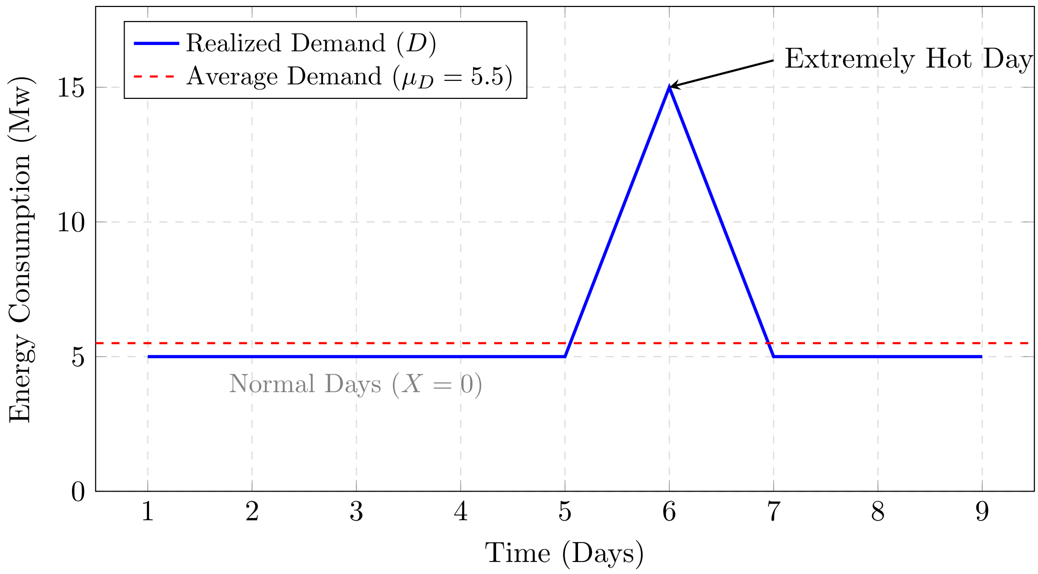


Figure 1.1: Energy consumption over time. The spike represents a rare event where temperature (X) affects the demand (D).

1.3 Demand Evolution over Time

So far, we have presented a static view of the demand as random variable. While useful as a first order tool of analysis, it is easy to imagine that the demand of many products is in fact time-dependent: Ice cream sales peak in the summer, airline traffic peaks around holidays and restaurants demand has modes around lunch and dinner. Similar to last example of the previous section, averaging without side information can generate misleading estimates for the demand. Therefore, if the demand is time-dependent it makes sense to include time in its modeling.

Time is represented as a variable $t = 0, 1, 2, \ldots$ where each value of t represents a moment in time. The unit of time can be taken as seconds, hours, days or years. Depending on the context of the specific supply chain decision: For example, in analysing the demand behaviour in an online store, it makes sense to have relatively small units of time. Conversely, when planning capacities or resources for facilities, it makes sense to consider the demand over years. Later it will also become evident that there is a trade-off between the historical amount of data available and the quality of the estimation of the demand.

Incorporating time into the demand estimation leads to indexing the demand random variable by time as D_t, denoting the random demand at time t. In most scenarios, we have access to the demand up to some point in time in the past. For example, suppose that you are given the monthly historical demand for the last 5 years. Chronologically, these are 60 datapoints $d_0, d_1, d_2, \cdots, d_{59}$. Data like this can be denoted compactly as $\{d_s\}_{s=0}^{t}$ (imagine a spreadsheet with a single column and a row for each entry of the demand at every date). The distinction between the capital and lowercase symbols for the demand denote the random variable D_t and certain outcome for it d_t. This is important, as even though the past is a realization of the demand, the outcome was a result of the probabilistic nature of the demand (for example, a month of low demand could be caused by an unexpected shock, and it might provide information for future outcomes of the demand).

Random variables that evolve over time are referred as *Stochastic Processes*. We present a couple of examples to get a feel for stochastic processes. One of the simplest models is the following:

$$D_{t+1} = \alpha + \beta D_t + V_t, \tag{1.6}$$

We call this the *Base Model* as it will allow us to model some common behaviours of the demand over time. In this model $\alpha, \beta \in \mathbb{R}$ are two constants while V_t is an independent random variable. In plain words, the equation says that the demand at time $t+1$ is equal to a constant α plus a term multiplying the previous period demand D_t, plus a random quantity V_t (normally this quantity expresses what's commonly referred as an error or simply unexplained variation of the demand). This model is related to the previous one (independent of time) when $\beta = 0$[4].

To see the dependence over time, assume that at time 0, there is an initial demand $D_0 \geq 0$. $D_1 = \alpha + \beta D_0 + V_0$, and $D_2 = \alpha + \beta D_1 + V_1 = \alpha + \beta(\alpha + \beta D_0 + V_0) + V_1 = \alpha + \beta\alpha + V_1 + \alpha V_0 + \beta^2 D_0$. Continuing this iterative process leads to the following expression:

$$D_{t+1} = \sum_{j=0}^{t-1} (\alpha + V_{t-j})\beta^j + \beta^t D_0, \tag{1.7}$$

Assuming V_t for $t = 0, 1, \ldots$ has identical and independent distribution with $\mathsf{E}(V_t) = 0$. Taking expectations yields the following:

$$\mathsf{E}(D_{t+1}) = \alpha \frac{1-\beta^t}{1-\beta} + \beta^t \mathsf{E}(D_0), \tag{1.8}$$

The first term of the right hand side comes from the finite sum of a geometric series[5]. Then,

[4] If V_t is a random variable so is $\alpha + V_t$. In fact, constants are random variables with their probability mass concentrated at a single value.

[5] Take $s_t = 1 + \beta + \cdots + \beta^t$ and $\beta s_t = \beta + \cdots + \beta^{t+1}$, subtracting the two equations yields the result shown.

the expected value of the demand depends on the expected value of the initial demand, plus a term that converges to $\alpha/(1-\beta)$ at t goes to ∞. Moreover, this convergence only occurs if $|\beta| < 1$. If $|\beta| > 1$ the demand explodes as time goes by (think why this is the case). When $\beta = 1$ with $\mathsf{P}(V_t = 1) = \mathsf{P}(V_t = -1) = 1/2$. The model retrieved is called a *Random Walk*. Among other interesting properties, random walks drift away indefinitely to $\pm\infty$ and can also be negative. See Figure 1.2 to see the different models. With a small modification, of taking $\ln(D_t)$ instead of D_t a *Geometric Random Walk* can be used to model the demand.

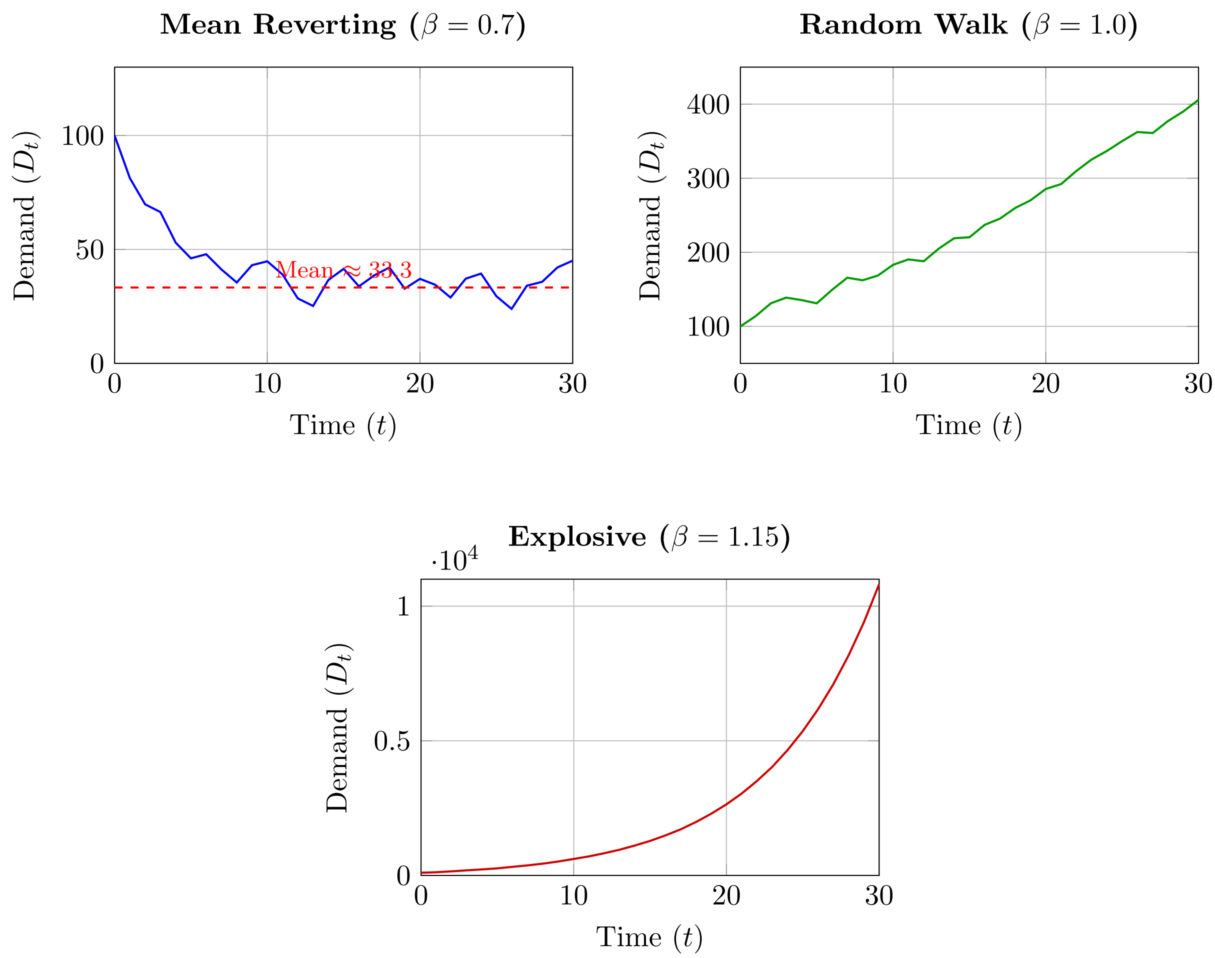


Figure 1.2: Stochastic demand simulations ($\alpha = 10, D_0 = 100, V_t \sim \mathrm{U}(-15, 15)$) illustrating the impact of β. The **Mean Reverting** ($\beta = 0.7$) case centers around $\alpha/(1-\beta)$, the **Random Walk** ($\beta = 1$) exhibits non-stationary drift, and the **Explosive** ($\beta = 1.15$) case follows a steep exponential trajectory.

1.3.1 Modeling Seasonality and Time-varying Factors

Another common feature of the demand is *seasonality*. Seasonality is defined as the fluctuation of the demand caused by the passage of time or certain time-dependent events (such as holidays or weather variations). Think of holidays and demand that changes due to weather or temperature.

The most common way to model this is to follow the method in the weather example of the previous section, by adding an extra variable codifying the side-information of interest. In the case of seasonality, this can be achieved by adding binary variables that are equal to 1 when t is equal to that time. For example, think of daily data where we can to capture day of week effect on the demand (think of a restaurant or retail business). A way to achieve this, is to modify the α term in Equation (1.6) as:

$$\alpha_t = \sum_{d=0}^{s-1} \mathbf{1}\{t \bmod s = d\}\gamma_d, \tag{1.9}$$

In this case, the indicator $\mathbf{1}\{t \bmod s = d\}$ is only one every s observations. In the days of the week example, $s = 7$ as there are 7 days in a week. Putting this expression back in the base model yields:

$$D_{t+1} = \sum_{d=0}^{s-1} \mathbf{1}\{t \bmod s = d\}\gamma_d + \beta D_t + V_t, \tag{1.10}$$

The model is quite similar to the original one (that had only one constant term α while this one has a constant term for every day of the week). In general, we can also include time-varying factors, such as seasons (also encoded in indicators or numerical variables. A clean way to include these factors is to flatten these variables into a vector h_t. For example, letting $h_t = (1, t)'$ retrieves a model with a constant plus a linear trend allows to express the original α_t term as $\alpha_t = Ah_t$ where A in this case is a vector of parameters with the same dimension as the number of time-varying variables, that is $A \in \mathbb{R}^{|h_t|}$. This allows to re-write the base model as:

$$D_{t+1} = \alpha_t + \beta D_t + V_t = Ah_t + \beta D_t + V_t, \tag{1.11}$$

This instance of the base model allows for arbitrary specifications of time-dependent variables. See Figure 1.3 for an example of the decomposition of the demand into different components.

1.3.2 Other Modeling Specifications

It is easy to imagine that a similar strategy can be employed to add arbitrary random variables or other stochastic processes X_t of variables that are related to the demand D_t. In general, one can imagine that any arbitrary real-valued function can be calibrated to solve the following problem:

$$\min_g \mathsf{E}[(g(D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots) - D_{t+j})^2]. \tag{1.12}$$

It should be evident that this is exactly the same problem in Section [1.2.2] where the optimal solution is $g(D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots) = \mathsf{E}(D_{t+j} | D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots)$. Then, speaking strictly from an optimization point of view, what matters is that the family of functions g can

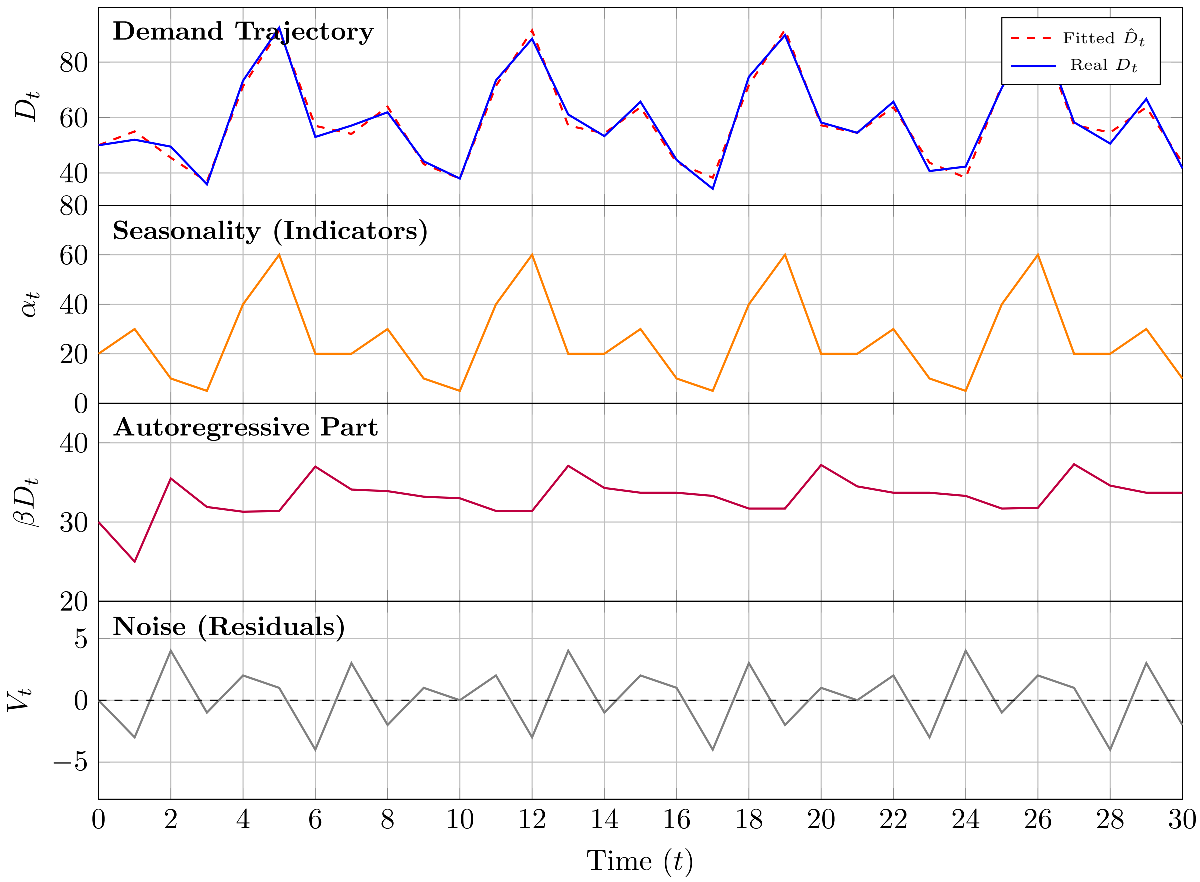


Figure 1.3: Demand decomposition into different components: Seasonal, autoregressive, and noise. Corresponding to each term in the specification of the demand.

express the conditional expectation of the demand D_{t+j} as a function of the other variables. Whether the model is a simple linear regression or a sophisticated neural network, what matters is that it can capture the shape of the conditional expectation. As we will see next, there is another tension related to the fact that in general the probabilistic distribution of the demand is unknown and only a finite sample of the variables has to be used for estimation.

1.4 Demand Estimation in Practice

So far, we have built a methodological theory to study the evolution of the demand over time. The core message is that the best possible guess for the demand is the conditional expectation of the demand taking into account information provided by other relevant variables. In practice, the situation is significantly different: At best there are limited observations of the historical demand (or no information at all in the worst case). The side-variables that affect the demand are unknown or unobservable and in many cases they are endogenous (competitors set prices reactively to their own demand and compete against each other for market domination). The latter situation is more complex and will be studied in a later chapter. In this section, we focus on the case where we have limited data and want to predict the demand in the *short-term*. Here, short-term is used loosely as the foreseeable future.

The main estimation problem for the demand is that its true probabilistic distribution is unknown. Then, the goal is to learn the demand from *sample* observations. Suppose, a historical yearly demand of salmon in a country. This demand should be stable and more or less proportional to the population of the country, to account for the population fluctuation, consider the *per capita* demand. This historical demand is represented by the observations $\{d_s\}_{s=1}^t$. Estimating the average demand from samples is calculated as:

$$\mu_D(t) = \frac{1}{t}\sum_{s=1}^{t} d_s, \tag{1.13}$$

This average converges to a stable (constant value) in various cases: For example, when the demand samples are *independent and identically distributed*, or i.i.d. Meaning that they come from the same probabilistic distribution, and the previous values do not affect the probability of new outcomes of the demand. One can see getting observations of the demand as sampling the outcome of an experiment repeatedly and measuring the demand after each trial.

It is important to determine how reliable this estimate is, especially when managing a supply chain as making resource allocation decisions in the future can be extremely costly. Probability theory has two answers regarding how trusted this estimator can be: The *Law of Large Numbers* and the *Central Limit Theorem*.

The Law of Large Numbers (LLN) states that for a large enough t, the average demand (sampled i.i.d. with finite mean and variance) $\mu_D(t)$ converges to the true average value μ_D. That is, $\mu_D(t) \to \mu_D$ as $t \to \infty$. As mentioned before, this is important because there is a probabilistic guarantee that sampling the demand i.i.d. and averaging its values over time recovers the true average demand.

The Central Limit Theorem (CLT) goes a step further and provides a probability distribution for the randomness of the demand. Given an i.i.d. sample of the demand $\{d_s\}_{s=1}^t$. The random variable defined by $\frac{\sqrt{t}(\mu_D(t)-\mu_D)}{\sigma_D}$ converges in distribution to a random variable Z distributed as a standard normal random variable[6].

Example 1.2 (Rice Storage Levels). The government of Japan asks you to conduct a study to determine what level of rice reserves they should hold. Their first question is how many years t of data they should look at to get a 95% confidence estimate of the average demand μ_D that is within 10% of the true value, using the sample estimate $\mu_D(t)$. Suppose you are told that the standard deviation of the demand σ_D is 30% of the mean μ_D. That is, $\sigma_D = \mu_D 30\%$.

[6] A standard normal random variable has probability density function (pdf) $f(x) = e^{-x^2/2}/\sqrt{2\pi}$. A pdf is the continuous analogous to the pmf in the sense that for a random variable X with density $f(x)$ the probability $\mathsf{P}[X \in (x, x+\delta_x)] = f(x)\delta_x$ for a small $\delta_x > 0$.

Using the CLT we get that the sample average demand is equal in distribution to:

$$\mu_D(t) \overset{\mathrm{d}}{=} \frac{\sigma_D}{\sqrt{t}} Z + \mu_D, \tag{1.14}$$

Dividing by μ_D yields the percentage error $\frac{\sigma_D}{\mu_D\sqrt{t}}Z + 1$. Where $\frac{\sigma_D}{\mu_D\sqrt{t}}Z$ is the error in terms of the demand, which can be positive or negative. Then, for this expression to within 10% of the true value with 95% confidence we need the following to be true:

$$\mathsf{P}\left(\left|\frac{\sigma_D}{\mu_D\sqrt{t}}Z\right| \leq 10\%\right) = 95\%, \tag{1.15}$$

Using the fact that $\sigma_D/\mu_D = 30\%$, the probability can be rearranged to $\mathsf{P}(|Z| \leq \sqrt{t}/3) = 1 - \mathsf{P}(|Z| > \sqrt{t}/3) = 95\%$ or $\mathsf{P}(Z > \sqrt{t}/3) = 5\%/2 = 2.5\%$. That is, the tail cdf of the standard normal distribution $\bar{\Phi}(\sqrt{t}/3) = 2.5\%$. Applying the inverse[7] tail cdf yields:

$$\sqrt{t}/3 = \bar{\Phi}^{-1}(2.5\%) = 1.96. \tag{1.16}$$

With this, we get our answer that the number of observations t needed for the sample average to be within 10% of the mean is at least $t = (3(1.96))^2 = 34.5$ years.

In general, the above example is the best possible case. In probability textbooks, a commonly encountered rule is that the sample size for estimation should be at least 30 observations. As both the LLN and the CLT relied on the i.i.d. property of the samples for convergence.

To illustrate this, consider a random demand that follows the following demand process $D_{t+1} = D_t + c + \sigma Z$, with $D_0 = 0$. Note that this process is an instance of the base model with $\alpha = c$, $\beta = 1$ and $V_t = \sigma Z$. In this case, $c + \sigma Z$ is called a normal random variable with mean c and variance σ^2 (likewise, a normal random variable can be transformed in a standard normal by subtracting its mean and dividing by σ. This operation is useful to make computations always in terms of standard normal random variables).

The demand process is equal to $D_t = ct + \sigma \sum_{i=1}^{t} Z_t$. The expected value of the demand $\mathsf{E}(D_t) = tc$, since each independent standard normal Z_t has mean 0. Likewise, the sample mean (extracted from historical observations of the demand) will be approximately this expected value, that is, $\mu_D(t) \approx tc$. As t increases indefinitely, so will the sample mean, that is, $\mu_D(t) \to \infty$ as $t \to \infty$. So, how can we estimate the demand in these cases?

Most textbooks in Time Series spend a significant amount of their exposition in studying *stationary* stochastic processes. That is, processes that converge to a stable mean in the long-term and do not explode indefinitely. The textbook solution is to apply a transformation to the original series, such that the transformation is stationary. In the previous example, a solution

[7] An inverse function $f^{-1}(x)$ is a function such that $f^{-1}(f(x)) = x$. For example, the inverse function of $f(x) = e^x$ is $f^{-1}(x) = \ln(x)$.

is to consider a new series taking the differences of the demand. For example, consider the difference process $\Delta_t = D_t - D_{t-1} = c + \sigma Z$, which as discussed previously is a normal random variable with mean c and variance σ^2. The sample mean of historical observations of Δ_t should be the same as sampling standard normal observations.

1.4.1 Data-driven Demand Estimation

In practice we cannot expect to know beforehand the actual transformation that makes the demand stationary, since we only have access to a historical demand series $\{d_s\}_{s=1}^t = \{d_1, d_2, \ldots, d_t\}$ that is sampled from an unknown stochastic process (with unknown mean, variance, seasonality and functional form).

The strategy that we will implement consists of the following steps:

1. Propose a functional form rich enough to capture the demand within a well-defined family of functional forms.

2. Calibrate the parameters of the function that fit the demand closest via optimization.

3. Evaluate the performance of the proposed function out-of-sample.

For Step 1, we can use some of the modeling techniques that we have discussed in Section 1.3. For example, we can use for example the model presented in the previous section, $\hat{D}_{t+1} = \hat{A}h_t + \hat{\beta}D_t$. Here, the hats above the expression denote quantities that are calibrated from data. Recalling the definition of conditional expectation in Section 1.2.2, a reasonable approach to calculate the parameters for the demand is to solve the following optimization problem:

$$\min_{\hat{A},\hat{\beta}} \mathsf{E}[(\hat{D}_t(\hat{A}, \hat{\beta}) - D_t)^2], \tag{1.17}$$

This expression can be seen as an instance of the conditional expectation defined as the best predictor for a random variable. The solution of this problem, is in fact an approximation of the conditional expectation of the demand, that is, $\hat{D}_t(A^*, \beta^*) \approx \mathsf{E}[D_t | D_{t-1}, \ldots]$. The quality of the approximation depends crucially on two factors: That the functional form $\hat{D}_t$ allows to approximate the conditional expectation. And also, on the sample size t of historical observations for the demand. Moreover, note that when the true parameters of the demand A and β are used, the difference $\hat{D}_t(A, \beta) - D_t \overset{\mathrm{d}}{=} V_t$. In general, $\mathsf{E}(V_t)$ should be equal to 0 (in terms of the base model, if V_t has non-zero mean c, define new variables $\alpha' = \alpha + c$ and $V_t' = V_t - c$, then $\mathsf{E}(V_t') = 0$). Therefore, this gives us another clue that $\mathsf{E}(\hat{D}_t(A^*, \beta^*) - D_t)$ should be approximately 0 when solving the optimization problem of fitting the demand.

To solve the optimization problem in Step 2, we solve the data-driven analogous of the problem in (1.17) with a sample of demand observations $\{d_s\}_{s=0}^t$ (in this case with the particular choice of demand function in Step 1):

$$\min_{\hat{A},\hat{\beta}} \frac{1}{t}\sum_{s=1}^{t}(\hat{A}h_{s-1}+\hat{\beta}d_{s-1}-d_s)^2, \tag{1.18}$$

This problem is a convex optimization program with a unique solution, that can be solved with any off-the-shelf quadratic optimization solver[8].

It is appealing to use a family of functions that can express very rich functional forms (for example, for a closed interval it is well-known that a Fourier basis can approximate any continuous function). The formalism would be to express the fitted demand $\hat{D}_t(\theta)$ as a function of parameters θ, for example, the weights of a deep neural network, or the collection of splits in a random forest. The approach would be the same, that is, to solve the optimization problem $\min_\theta \mathsf{E}[(\hat{D}_t(\theta)-D_t)^2]$.

The issue with this approach is that if the family of functions is too rich, the sample minimization would be too optimistic (potentially 0 error as $\hat{D}_t(\theta^*)=d_t$, for all observations in the sample $\{d_s\}_{s=1}^t$) and the performance for future predictions could be arbitrarily off. The two main reasons for this phenomenon are: First, in practice we only have access to *finite* samples that might not represent the full geometry of the conditional distribution (a well-known phenomenon in statistical analysis). Second, real-life stochastic process are in general changing in distribution over time (a so-called *doubly stochastic process*, where the original process changes distribution based on another unknown random variable). In the context of demand estimation, this could be a shock to the demand or a technological change. For example, the demand for horse carriages catastrophically decreased with the invention and mass production of mechanical engine cars (think that the measure of power of engines is called *horsepower*) in the early twentieth century. Likewise, the introduction of new competitors, or shocks to the demand or supply can change the trajectory of the demand.

While this problem cannot be fully mitigated, if there is enough data we can use what we have learned to come up with a partial solution: For the first problem of over-fitting the demand, we can sub-divide our full sample of historical observation into multiple sub-samples where by setting up an experiment, we can use the LLN and the CLT to get an idea of the performance of our demand estimation strategy out-of-sample. For the second problem, imagine that distribution changes at random with a throw of a coin with small probability p, then, the time τ in between changes in distribution is in average $\mathsf{E}(\tau)=1/p$. Then, if instead of training our model with all of the historical observations, we train the model with k observations, such that $k<\mathsf{E}(\tau)$, with probability $(1-p)^k \approx 1$ we are in the current distribution of the demand (another approach would be to weight the observations with a decaying weight). To see the point of using k observations imagine a sudden financial crisis that impacts the demand of luxury

[8] Linear least squares is an instance of Quadratic Programming. Note that in matrix form linear least squares has the objective $||\beta X-y||_2^2 = 2(\beta XX^T\beta/2-\beta Xy^T+yy^T/2)$ which is a quadratic program by letting $Q=XX^T/2$ and $c^T=-Xy^T$. Then, $\min_\beta ||\beta X-y||_2^2 = 2\min_\beta(\beta Q\beta^T+\beta c^T)+yy^T$.

goods, such as designer bags. Using the whole historical demand will make the estimates to continue growing if the historical data includes most of the period before the crisis. Using the last k observation makes the estimates adjust faster to the new distribution of the demand. For another example, imagine a positive shock to the demand, in Figure 1.4 a fixed model cannot update to changes in the distribution of the demand.

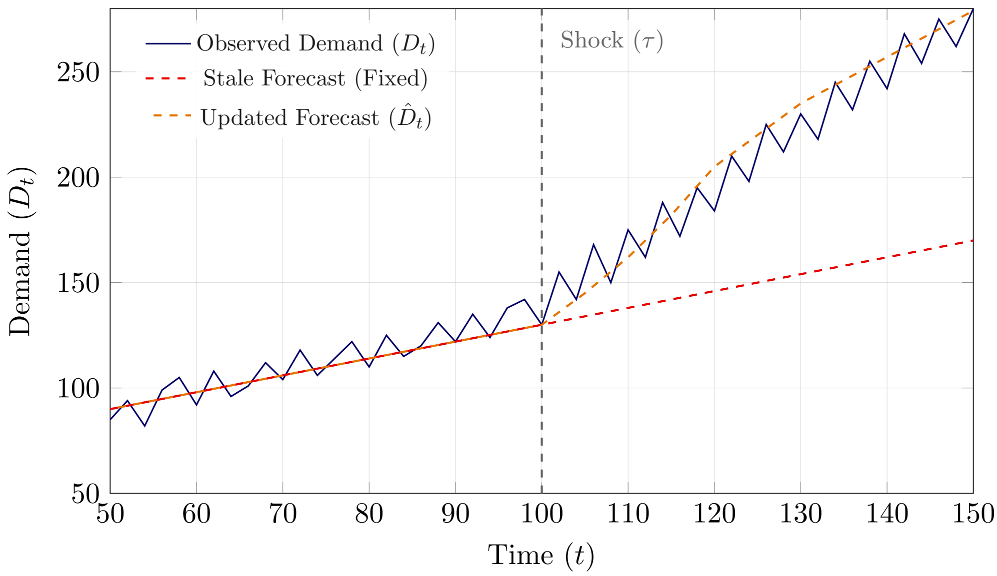


Figure 1.4: The impact of structural shocks on forecast divergence. Prior to the shock ($t < 100$), both models track the same pre-shock parameters; an alternating dash pattern (red/orange) is used to display both distinct forecasting paths. After the shock (τ), the stale model (red) fails to account for the change, while the updated model (orange) recalibrates and captures the new demand trajectory.

Then, for Step 3 we take the following approach: Given historical observations of the demand $\{d_s\}_{s=1}^{T}$, a estimation window size k and a prediction horizon j. Define the sample optimization problem as $\ell(t, t') = 1/(t' - (t+1)) \sum_{s=t+1}^{t'} (Ah_{s-1} + \beta d_{s-1} - d_s)^2$. Likewise, define the absolute error $|\varepsilon_{t+j}(A, \beta)| = |\hat{D}_{t+j}(A, \beta) - d_{t+j}|$. The strategy is going to be to estimate multiple observations of the error out-of-sample, and average them as a measure of how the model performs out of sample. The pseudo-code routine to perform the *backtest* can be written as:

With this procedure, given two models (say the linear one that we have been exploring) and a more sophisticated one like a Deep Neural Network model. The estimates of the mean absolute error μ_ε can be used to compare the performance of the model out-of-sample. This procedure is commonly used in finance to backtest algorithms. The key concept is that the measurement of the error is done outside of the training data of the model, thus, mimicking the behavior of the model in real life. Likewise, the choosing of the training window size k can be evaluated with running this procedure multiple times with different sizes of k. Of course, to make the comparison fair, some exclusion of data might be needed as a larger window size

Algorithm 1 *Backtesting Demand Prediction*

Input: Historical observations for the demand $d_1, \dots, d_T$. A training window k, and a prediction horizon j. And a model specification for the demand $\hat{D}_{t+j}(\cdot)$

Output: The average out-of-sample absolute error $\mathsf{E}(|\varepsilon_{t+j}|)$

procedure BACKTEST($\{d_s\}_{s=1}^{T}, k, j$)

  $\mu_{|\varepsilon|} = 0$

  for $t = k$ **to** $T - j$ **do**

    $A^*, \beta^* \leftarrow \text{argmin } \ell(t-k+1, t)$

    $|\varepsilon_{t+j}(A^*, \beta^*)| = |\hat{D}_{t+j}(A^*, \beta^*) - d_{t+j}|$

    $\mu_{|\varepsilon|} = \mu_{|\varepsilon|} + |\varepsilon_{t+j}(A^*, \beta^*)|/(T-k-j)$

  end for

  return $\mu_{|\varepsilon|}$

end procedure

needs more observations, and the number of observations for the error ε_{t+j} will differ depending on the window size. Lastly, this procedure can be easily modified to estimate the variance of the error or other metrics of interest.

The errors are themselves objects of interest. As discussed previously $V_{t+j} \stackrel{\mathrm{d}}{=} \hat{D}_{t+j}(A, \beta) - D_{t+j} = \varepsilon_{t+j}(A, \beta)$. Then, the mean $\mu_{\varepsilon(t+j)}$ and the variance of the errors $\sigma^2_{\varepsilon(t+j)}$ can be used to approximate the error of predictions. For example, approximating $V_{t+j} = \mu_{\varepsilon(t+j)} + \sigma_{\varepsilon(t+j)} Z$ allows to construct confidence intervals for the demand. For example, the probability that the forecasted demand at time $t + j$ exceeds the quantity x is equal to:

$$\mathsf{P}(D_{t+j} > x) = \mathsf{P}(\hat{D}_{t+j}(A, \beta) + V_{t+j} > x) = \mathsf{P}\left(Z > \frac{x - \hat{D}_{t+j}(A, \beta) - \mu_{\varepsilon(t+j)}}{\sigma_{\varepsilon(t+j)}}\right). \quad (1.19)$$

Which is the well-known tail cdf of a standard normal distribution $\bar{\Phi}\left(\frac{x - \hat{D}_{t+j}(A,\beta) - \mu_{\varepsilon(t+j)}}{\sigma_{\varepsilon(t+j)}}\right)$. It is important to note that the prediction horizon j is crucial, because in general the longer the prediction horizon, the larger the errors. That being said, there is no free lunch in prediction since the estimation of the mean error and its variance is in itself a random variable. Moreover, there is an implicit assumption that the distribution of the stochastic process is in some sense stationary or fixed within a time interval. In the real-world there is no guarantee that this is the case.

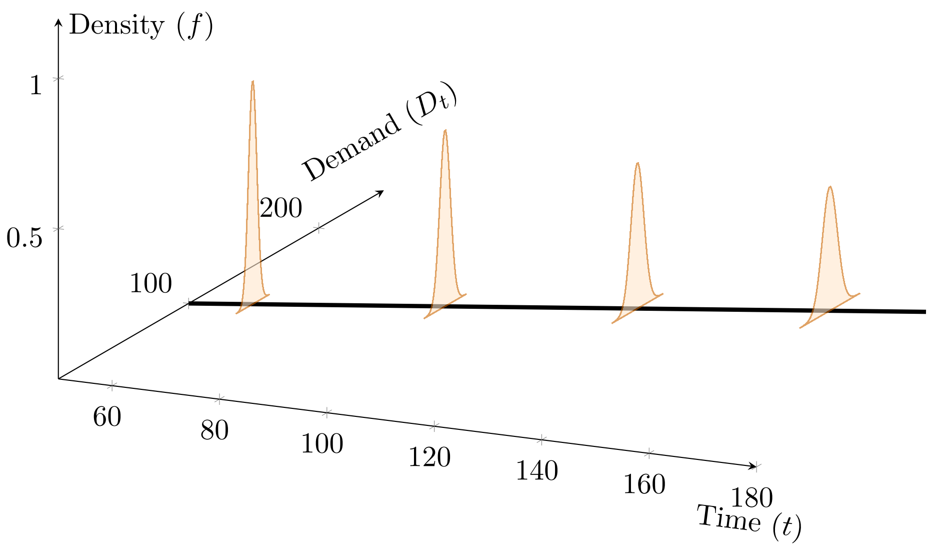


Figure 1.5: The prediction trajectory $\hat{D}_{t+j}$ is plotted against Time (t), with the vertical Z-axis representing the probability density function. The increasing spread of the normal distributions illustrates the growing uncertainty $\sigma_{\varepsilon(t+j)}$ over the prediction horizon.

Chapter 2

Inventory Management Models

2.1 Introduction

Even with a good grasp of the magnitude and direction of the randomness affecting the future demand, there is always a decision behind predicting the future demand. Allocating resources to match random demand is a difficult problem. The difficulty lies in that resources are often fixed and not flexible: Determining the production capacity of a building, ordering a fixed amount of materials to manufacture products, or scheduling workers ahead of time to service a facility or store. What all these decision have in common is that there is an inherent trade-off between allocating too many resources or allocating too little. On one hand, allocating too many resources is costly and inefficient, and on the other, allocating too little, leads to congestion and lost sales. Both are undesirable outcomes that have to be balanced in an optimal manner. In this chapter, we present strategies to balance optimally these trade-offs in a data-driven way.

2.2 The Newsvendor Model

Following our exposition of characterizing the demand as a Stochastic Process $\{D_t\}_{t=0}^{\infty}$, one of the most popular production models is known as the Newsvendor Model [2]. In this model, a production or inventory decision quantity q has to be taken at time $t = 0$ to satisfy a random demand D_T at horizon time T. The goal of the model is to determine what is the optimal quantity of product that should be produced/ordered. Product is sold at a price p with a unit cost c. The objective function of the seller is their expected profit, defined as:

$$\max_q \mathsf{E}[(D_T \wedge q)p - qc], \tag{2.1}$$

The revenue of the firm is the minimum operator (denoted by $a \wedge b = \min(a, b)$) between the production/inventory quantity q and the random demand D_T multiplied by the price p. As the production cost of each unit is c, the total variable cost is qc. Subtracting the cost from the revenue gives the profit (note that fixed costs do not affect the production quantity as they do not depend on q).

The first step to solve this problem is to note for any numbers $a, b \in \mathbb{R}$ we have $a \wedge b = a - (a-b)^+$ where $(x)^+ = \max(x, 0)$. With this fact, the profit can be re-written as $(D_T \wedge q)p - qc = [q - (q - D_T)^+]p - qc = q(p-c) - p(q - D_T)^+$. Then, the profit can be re-written as:

$$\max_q q(p-c) - p\mathsf{E}(q - D_T)^+, \tag{2.2}$$

To solve the problem we need to calculate what is the critical point q^* such that the first derivative of the profit is equal to 0. We also need to check that the function is concave, as this will ensure that the critical point of the unique optimal solution maximizing the profit. Taking derivate and equating to 0, yields $\frac{\partial \mathsf{E}(q-D_T)^+}{\partial q} = \frac{p-c}{p}$. We can use our definition of expected value (in Section 1.2) to write $\frac{\partial \mathsf{E}(q-D_T)^+}{\partial q}$ as follows:

$$\frac{\partial}{\partial q}\sum_{i=0}^{\infty}(q-i)^+ p_i = \frac{\partial}{\partial q}\sum_{i=0}^{\infty}(q-i)p_i \mathbf{1}\{i \le q\} = \frac{\partial}{\partial q}\sum_{i=0}^{q}(q-i)p_i = \mathsf{P}(D_t \le q) = F_{D_T}(q). \tag{2.3}$$

The first equality is just a re-writting of the event $(x)^+ = x\mathbf{1}\{x \ge 0\}$. The second one just changes the limit of the sumation where the inner term is non-negative. The third equality exploits the fact that the derivative is a linear operator[1] and that the derivative of qp_i is p_i with respect to q. The second derivative is equal to 0, thus making the profit function concave and ensuring that the critical point in indeed the quantity that maximizes the profit. The optimal quantity is:

$$q^* = F_{D_T}^{-1}\left(\frac{p-c}{p}\right), \tag{2.4}$$

$F_{D_T}^{-1}$ denotes the inverse cdf of the distribution of the demand. Connecting this to the demand distribution Section 1, for example by taking the data-driven estimate of the demand as the approximation $D_T = \hat{D}_T(\theta) + V_T \stackrel{\text{d}}{=} \mu + \sigma Z$ with $\mu = \hat{D}_T(\theta)$ (in general, an arbitrary functional form for the demand. With $\theta = A, \beta$ we can take this as an instance the *base model*) and $\sigma = \sqrt{\mathsf{Var}(V_T)}$. With this approximation, the optimal quantity becomes:

$$q^* = \mu + \Phi^{-1}\left(\frac{p-c}{p}\right)\sigma. \tag{2.5}$$

Where Φ^{-1} is the inverse cdf of the normal distribution. The interpretation of this expression is that the optimal quantity q^* is the predicted value from the model $\mu = \hat{D}_T(\theta)$, plus a buffer proportional to the standard deviation of the demand $\sigma = \sqrt{\mathsf{Var}(V_T)}$. The multiplier $\Phi^{-1}\left(\frac{p-c}{p}\right)$ balances how aggressive the provisioning of inventory should be. First, the price p is normally higher than the cost c, that is, $p > c$. To get a feel for the multiplier, suppose the price is

[1] The derivative of a sum is the sum of the derivatives of its terms.

3 times the cost, or $p = 3c$ (a typical pricing strategy in the absence of information), then $(p - c)/p = 2/3$. Evaluating the inverse at this value gives $\Phi^{-1}(2/3) \approx 43\%$, meaning that the extra buffer above the predicted demand μ is 43% of its standard deviation σ. The larger the difference between the price and cost is $p - c$, the greater the amount of provisioning should be (because the loss of profit from a sale lost is orders of magnitude higher than the cost). As we will see next, we can modify this model to more complex situations where there are costs associated with unsold product. See Figure 2.1 for a visual representation.

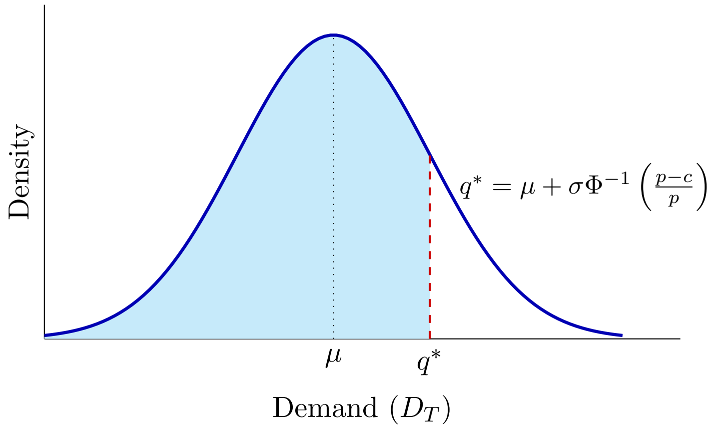


Figure 2.1: Optimal inventory level q^* in the Newsvendor Model. The shaded area represents the probability of meeting all demand, corresponding to the critical ratio $(p - c)/p$.

Example 2.1 (Avocado Toast Restaurant). A national restaurant chain serves a signature avocado toast dish in their menu. The restaurant procures avocados from a wholesale producer and can purchase avocados at a price $c = \$0.68$ per unit. Every avocado dish is sold at a price $p = \$12$, other variable costs add up to $c_a = \$3$ per dish (other ingredients and labor). Every unused avocado has to be disposed at a cost of $c_d = \$0.20$. The distribution of the demand at time T is estimated as a normal random variable with mean $\mu = 70$ and standard deviation $\sigma = 10$ dishes for a specific location of the restaurant. Also, when customers arrive and the avocado toast is sold-out, with probability $p_r = 1\%$ they never come back (imagine a binary random variable B_r with $\mathsf{P}(B_r = 1) = p_r = 1 - \mathsf{P}(B_r = 0)$, independent of the demand). The average lifetime profit per customer is estimated to be $\ell = \$200$. What is the optimal quantity of avocados that maximizes the expected profit for this location? What happens when the lifetime profit per customer increases to $\ell = \$2,000$?

The expected profit for the restaurant can be written as:

$$\max_q \mathsf{E}[(D_T \wedge q)p - q(c + c_a) - (q - D_T)^+ c_d - (D_T - q)^+ B\ell], \tag{2.6}$$

Rearranging terms (noting $a \wedge b = a - (a-b)^+$ as in Equation (2.2)) leads to this expression:

$$\max_q q(p - c - c_a) - (c_d + p)\mathsf{E}(q - D_T)^+ - p_r \ell \mathsf{E}(D_T - q)^+, \tag{2.7}$$

Similar to the previous example, the strategy to solve the problem is to take derivatives with respect to q and equate to 0. The derivative $\frac{\partial \mathsf{E}(D_T - q)^+}{\partial q}$ is equal to $-\bar{F}_{D_T}(q) = F_{D_T}(q) - 1$ (the trick is the same as in Equation (2.3), noting that the expression is non zero when the demand is between q and ∞). Grouping terms leads to the following optimal demand quantity:

$$q^* = F_{D_T}^{-1}\left(\frac{p - c - c_a - p_r \ell}{p + c_d - p_r \ell}\right). \tag{2.8}$$

In the case of this example, this leads to $q^* = \mu + \sigma\Phi^{-1}\left(\frac{p-c-c_a-p_r\ell}{p+c_d-p_r\ell}\right) \approx 76.03$ avocados. When the lifetime profit per customer ℓ is \$2,000 the optimal quantity rises to $q^* \approx 81.72$ avocados.

2.2.1 Empirical Distributions

The Newsvendor model relies heavily on the accuracy of the inverse cdf $F_{D_T}^{-1}$. In real life, the true distribution of the demand is rarely known, and the estimation has to be done on the empirical distribution from a sample. The empirical distribution of the demand is constructed by first sorting observations of the demand in ascending order. A sample of i.i.d. observations of the demand $\{d_i\}_{i=1}^N$ (here, the big assumption is that these are all independently sampled observations of the demand D_T). Let $d_{(i)}$ be the i-th smallest element of the sample, such that $d_{(1)} \leq d_{(2)} \leq \cdots \leq d_{(N)}$. The *Empirical probability measure* assigns each sample a equal probability $1/n$. The sample or empirical mean is then defined as $\frac{1}{n}\sum_{i=1}^n d_i$ which by the LLN converges to the true mean as we discussed in the previous chapter. Likewise, the empirical cdf is defined as $\hat{F}_n(q) = i/n$ for $i = \text{argmax}\{i : d_{(i)} \leq q\}$. Similarly, the empirical inverse cdf is defined as $\hat{F}_n^{-1}(p) = d_{(i)}$ for $i = \text{argmin}\{i : i \geq pn\}$. A natural question is how far off is the Newsvendor inventory quantity when using the empirical cdf in lieu of the real one.

A useful bound for the divergence of the empirical cdf and the actual one was developed in two papers [3, 4] and is commonly known as the *DKW bound* (due to the authors of the first paper, which was later refined by the second one). The bound is stated as follows:

$$\mathsf{P}\left(\sup_q |F(q) - \hat{F}_n(q)| > \lambda\right) \leq 2e^{-2n\lambda^2}, \tag{2.9}$$

What this bound states is that the probability that the deviation from the empirical and the actual cdf exceeds a value λ (for any value q) is less than $2e^{-2n\lambda^2}$. As n increases, this probability decays to 0. Meaning that as the sample size increases, the accuracy of the empirical cdf converges to the true one. Interestingly, we can use this inequality to build confidence intervals on the empirical cdf. For example, with probability $1-\alpha$ the true value of the cdf $F(q)$ is within $\hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}$ (hint: equate the right hand side of (2.9) to α) for any q.

Going back to our topic of interest, we want to know how far is the solution of the empirical Newsvendor model $\hat{q}_n^* = F_n^{-1}(\gamma)$ with respect to the real solution $q^* = F^{-1}(\gamma)$ (here, γ denotes the optimal constant in the Newsvendor model, for example $\gamma = \frac{p-c}{c}$ in the plain vanilla formulation). In Section 1 we emphasized that event calculus is a powerful problem solving tool. That is, to write complex events in terms of events that we know well probabilistically. In this case, we know about bounding the empirical cdf in Equation (2.9) and need to find the right event to express the relation to the empirical inverse cdf $F_n^{-1}(\gamma)$.

The optimal quantity $q^* = F^{-1}(\gamma) = \inf\{q : F(q) \geq \gamma\}$. Now we know that $F(q)$ is within $\hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}$ with probability at least $1-\alpha$. Putting these two facts together we get that $q^* \in \inf\{q : \hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)} \geq \gamma\}$ or $q^* \in F_n^{-1}\left(\gamma \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}\right)$ with probability $1-\alpha$. Note that this bound is not symmetric and not centered at $F_n^{-1}(\gamma) = q_n^*$. See Figure 2.2.

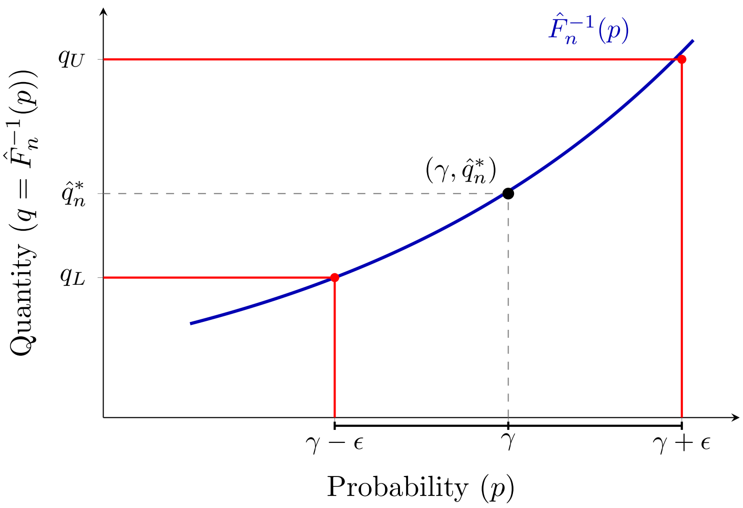


Figure 2.2: Mapping the DKW confidence interval. The symmetric interval $\gamma \pm \epsilon$ results in a non-symmetric confidence interval $[q_L, q_U]$ due to the nonlinearity of the inverse CDF.

Often, it is also a natural question what should be the average profit using a quantity $\hat{q}$ (for example, the optimal sample inventory q_n^*) given a series of n historical observations of the demand. The sample average profit $\mu_\pi(n)$ of the plain vanilla Newsvendor model is:

$$\mu_\pi(n) = \frac{1}{n}\sum_{i=1}^{n} \hat{q}(p-c) - p(\hat{q} - d_i)^+, \tag{2.10}$$

We know that because of the CLT, this mean will converge to the true average profit $\mu_\pi = \hat{q}(p-c) - p\mathsf{E}(\hat{q} - D_T)^+$. The variance of the profit is $\mathsf{Var}(p[(\hat{q} - D_T)^+]) = p^2\mathsf{Var}(\hat{q} - D_T)^+$, for any distribution of the demand D_T, this variance is at most $q^2/4$[2]. Then, we have the following probabilistic relationship between the sample and true profit:

$$\mu_\pi(n) \stackrel{\mathrm{d}}{=} \mu_\pi + \frac{p\sqrt{\mathsf{Var}(\hat{q} - D_T)^+}}{\sqrt{n}} Z \leq \mu_\pi + \frac{p\hat{q}}{2\sqrt{n}} Z, \tag{2.11}$$

[2]This is the so-called Popovicius's variance inequality. Let X be a random variable such that $\mathsf{P}(X \in [a,b]) = 1$, then, $\mathsf{Var}(X) \leq (b-a)^2/4$.

The first equality is the equality in distribution of the CLT. The second is the result of bounding the variance with the inequality above. The result says that for any distribution of the demand, in the i.i.d. case solving the plain vanilla Newsvendor problem converges to the true demand as n grows, and that the noise is bounded by the product of the price p and the quantity $\hat{q}$ divided by a factor of $2\sqrt{n}$. For example, to obtain a confidence interval with probability $1-\alpha$ of the true average profit μ_π by using quantity $\hat{q}$, we have that the error $|\mu_\pi - \mu_\pi(n)| \overset{\mathrm{d}}{=} \frac{p\hat{q}}{2\sqrt{n}}|Z|$. Then, we want to find the value ϵ such that $\mathsf{P}\left(|Z|p\hat{q}/(2\sqrt{n}) > \epsilon\right) = \alpha$, which is the same as $\mathsf{P}(Z > 2\epsilon\sqrt{n}/(p\hat{q})) = \alpha/2$. Applying the inverse tail cdf of the normal distribution, we get solving for ϵ:

$$\mu_\pi \in \mu_\pi(n) \pm \bar{\Phi}^{-1}(\alpha/2)\frac{p\hat{q}}{2\sqrt{n}} \text{ w.p. } 1-\alpha, \tag{2.12}$$

For example, choosing $\alpha = 5\%$, $\bar{\Phi}^{-1}(\alpha/2) = 1.96$, meaning that the true profit is with 95% probability within $1.96\frac{p\hat{q}}{2\sqrt{n}}$ of the sample profit in the plain vanilla Newsvendor model. This technique can also be used for all variations of the Newsvendor problem in the i.i.d. case, noting that the complement variance $\mathsf{Var}(D_T - \hat{q})^+$ is also bounded.

These bounds are very useful because the do not depend on the distribution of the demand and allow to infer the true performance of the model as a function of the sample size. We illustrate this in the following example.

Example 2.2 (Launching a new product). A supermarket chain is planning to release a new bottled orange juice. They perform a pilot of 30 days ($n = 30$), observing an average daily profit of \$1,500. The price of the juice is \$3.49 and they produce per store $\hat{q} = 100$. Assuming the distribution of the demand of juices is i.i.d. in the plain vanilla Newsvendor model. The shelf opportunity cost of the supermarket (the minimum profit that a product makes in that shelf area) is \$1,450 per day. With this data, the supermarket wants to conduct a hypothesis test to decide whether to include the orange juice permanently or continue the pilot. What's the probability that the daily profit is less than \$1,450?

With our analysis we know the true profit μ_π is bounded in distribution to $\mu_\pi(n) + \frac{p\hat{q}}{2\sqrt{n}}Z = \$1,500 + (\$349/2\sqrt{30})Z$. Then $\mathsf{P}(\$1,500 + (\$349/2\sqrt{30})Z < \$1,450)$ simplifies to $\mathsf{P}(Z < -\$50(2\sqrt{30})/\$349)$, which is $\mathsf{P}(Z < -1.56) = \Phi(-1.56) = 5.9\%$. Which means that for a 95% confidence one sided test, the pilot should continue (as the probability is above 5%).

2.3 Inventory Replenishment - The (s, S) rule

In the previous section, we proposed a model for optimizing a production quantity such that profit is maximized in a static setting. The methodology can be extended to optimize multiple products and locations. As well as optimizing production accross time. In this section, we deal with a common inventory policy taking into account the evolution of the demand over time.

In the previous chapter, we characterized the randomness of the demand as a Stochastic Process $\{D_t\}_{t=0}^{\infty}$. Assuming the distribution of the demand is known, define the inventory level at period t by I_t. The inventory can be defined as the quantity of a commodity or product that is stored to service a random demand. The simplest rule to replenish the inventory is known as the (s, S) rule. This rule works as follows: at the end of period t, if the inventory I_t is below a level s. An order (assumed to fulfill instantaneously) replenishes the inventory to a level S. Then, the inventory I_{t+1} is equal to $S - D_{t+1}$ when $I_t \leq s$, and it is $I_t - D_{t+1}$ otherwise (in the case when $I_t > s$). Clearly, the time between inventory replenishments is random and depends on the distribution of the demand.

Under the (s, S) rule, the inventory is in itself a random variable that depends on the demand. The conditional distribution of the inventory is given by:

$$\mathsf{P}(I_{t+1} = j | I_t = i) = \mathsf{P}(D_{t+1} = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j), \tag{2.13}$$

When the distribution of the demand is i.i.d., that is, $D_t \overset{\text{d}}{=} D$ for all t. We can express the inventory process I_t as a Markov Chain[3] with transition probabilities $p_{ij} = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$. Note that the inventory can take negative values as the demand can exceed the inventory quantity I_t. Conversely, under the (s, S) rule, the inventory never exceeds S by construction. See Figure 2.3.

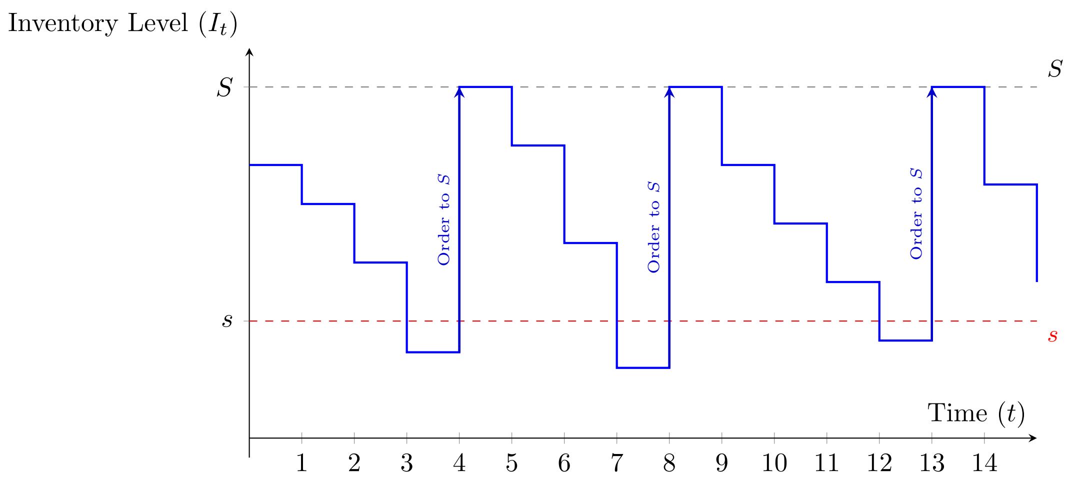


Figure 2.3: Inventory level I_t over time under the (s, S) policy. Replenishment is triggered the moment $I_t < s$, instantly bringing the stock back to the maximum level S.

With these elements we have enough building blocks to express problems of interest: For example, suppose that everytime there is a replenishment of inventory, there is a fixed cost K

[3]A Markov Chain is a special class of Stochastic Processes, where the distribution of the process X_t depends only on the realization of the process the previous period X_{t-1}.

plus a variable cost of c per unit. Likewise, suppose there is a holding cost of c_h per day the inventory is stored. Finally, when the inventory is negative (meaning that there was not enough inventory to service the demand), those sales are lost, a loss of a net revenue $\tilde{p}$ per unit. We can summarize these costs as a function $c(i)$ for each possible state of the inventory:

$$c(i) = \begin{cases} ic_h, & \text{if } s \leq i \leq S, \\ ic_h + K + (S-i)c, & \text{if } 0 \leq i < s, \\ K + Sc - i\tilde{p}, & \text{if } i < 0. \end{cases} \tag{2.14}$$

A Markov Chain transitions between states forever (even when it gets absorbed into a single state). When the transitions can be expressed with a fixed matrix P with elements p_{ij} denoting the probability that the chain at state i transitions to state j. The so-called limiting distribution is given by a vector π as:

$$\lim_{t\to\infty} P^t = \pi \quad \text{or} \quad \pi P = \pi, \tag{2.15}$$

Verbally, the vector π denotes the distribution of the states long-term. For example, $\pi(i) = \lim_{t\to\infty} \sum_{s=1}^{t} \mathsf{E}\mathbf{1}\{X_s = i\}/t$ shows what proportion of time (out of t) is spent in state i for a Markov Chain $\{X_t\}$.

In our case, this limiting distribution π is useful because we know we can express our inventory model as a Markov Chain with transition probabilities $p_{ij} = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$. Then, the long-term cost of the (s, S) inventory rule is given by:

$$\sum_{i=s-M}^{S} \pi(i)c(i), \tag{2.16}$$

Where M is given by a value such that $\mathsf{P}(D \leq M) = 1$. That is, the largest the demand can be (why is the lower limit of this Markov Chain $s - M$?). To compute the limiting distribution we just need to solve the system of equations $\pi P = \pi$ numerically. We spend the rest of this section discussing how to find actual values for (s, S). See Figure 2.4 for an example analysis.

2.3.1 Finding the (s, S) values in the i.i.d. case

The most direct approach to find a pair of values $0 \leq s < S$ that is optimal in the sense of minimizing is to enumerate all possible combinations of values of s, S and calculate the stationary distribution by solving $\pi P(s, S) = \pi$, where $P(s, S)$ is the transition matrix under the policy (s, S) (recall, the transition probabilities depend on both s and S, as $p_{ij}(s, S) = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$). We summarize the procedure in the following algorithm:

Algorithm 2 *Finding Optimal* (s^*, S^*) *Inventory Policy - i.i.d. case*

Input: A distribution of the demand $\mathsf{P}(D = i)$. A cost function $c_\theta(i)$ such as the one in Equation (2.14) with parameters θ. And upper limit of inventory capacity S_{max}, and for the demand M.
Output: The optimal inventory policy parameters (s^*, S^*).
procedure OPTIMALINVENTORY(F_D, c_θ, S_{max})
    $S, s^*, S^* = S_{max}$
    $c^*, \bar{c} = \infty$
    for $S = 1$ **to** S_{max} **do**
        for $s = 0$ **to** $S - 1$ **do**
            $P \leftarrow [p_{ij}] = [\mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)]$ for $i, j \in \{s - M, \cdots, S\}$.
            Solve π such that $\pi = \pi P$ and $|\pi| = 1$.
            Update and compute $\bar{c} \leftarrow c_\theta' \pi$.
            if $\bar{c} < c^*$ **then**
                Update $c^* \leftarrow \bar{c}, s^* \leftarrow s, S^* \leftarrow S$.
            end if
        end for
    end for
    return s^*, S^*, c^*.
end procedure

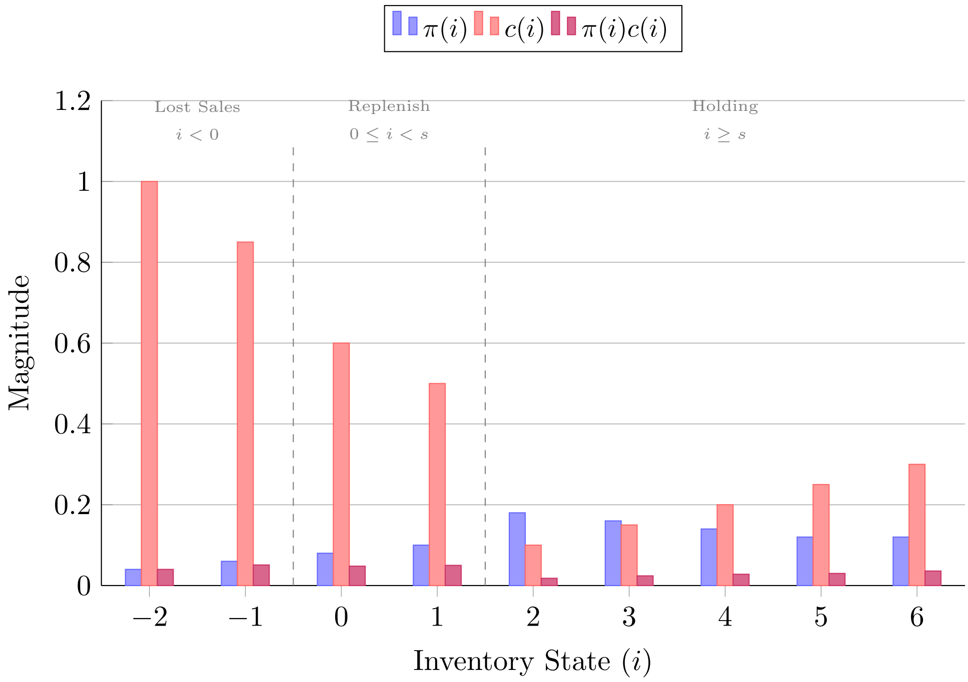


Figure 2.4: Long-term inventory cost analysis. The purple bars represent the weighted cost contribution of each state to the total expected cost $\sum \pi(i)c(i)$. Optimization aims to choose s and S such that the high-probability states (π) align with low-cost states (c).

Given the structure of the transitions of the inventory, this is not the most efficient algorithm to solve the inventory policy, but it is the easiest to implement as a computational routine. Next we examine the case to find policies for time-varying demands.

2.3.2 The time-varying demand optimal inventory

The key observation in the previous subsection was that the inventory process is a Markov Chain, that is, that the transition probabilities are time-homogeneous. As we explored in Section 1, demand is often seasonal and time-dependent. In this section, we present a useful technique to solve problems that have a time-dependent structure.

In the time-dependent setting, the demand distribution $F_{D_t}(i) = \mathsf{P}(D_t \le i)$ changes from period to period. Suppose we want to derive an inventory policy, that at each period tells us the optimal ordering level that minimizes some cost (for example, the cost function in Equation (2.14)). Given a time horizon of T periods in the future. The objective is to minimize a cost over the time horizon as:

$$\mathsf{E}\left[\sum_{t=1}^{T} c_t(I_t)\right], \tag{2.17}$$

The trajectory of the inventory I_t depends on the previous periods and is not time homogeneous, making the problem seemingly hard to solve. A useful technique to solve sequential problems is to look one step ahead (or behind) and try to define a recursive relation that allows to solve the problem going to the last period T (where the decision is normally trivial) and solve the problem *backwards*, in a recursive way. The easiest way to set up such relation is to define the *action* that can be taken separated from the *state* of the system, so that at each step the problem reduces to find the optimal action at that state in that time step. The state of the system in this case is already defined as the inventory level I_t, an integer (potentially negative) denoting the inventory amount at the end of period t. In the previous section, for the (s, S) policy, the action was coded in an indicator function (as $I_{t+1} = S\mathbf{1}\{I_t < s\} + I_t(1 - \mathbf{1}\{I_t < s\}) - D_{t+1}$ or $I_{t+1} = I_t + (S - I_t)\mathbf{1}\{I_t < s\} - D_{t+1}$). Inspired by this, we can express the state and the action of the system $I_{t+1} = j - D_{t+1}$ such that $j \geq I_t = i$. In this case, j represents the new level of replenishment of the inventory (or rather, the inventory at the beginning of $t+1$, before the demand D_{t+1} is subtracted). The strategy is then to write a function $\bar{C}_t(i)$ that represents the optimal expected cost from time t to T, that is, $\bar{C}_t(i) = \mathsf{E}\left[\sum_{s=t}^{T} c_s^*(I_s)|I_t = i\right]$, where c^* is the optimal cost of taking the best inventory actions. The strategy is then, to write down a one-step recursion that allows us to break down the problem to a single replenishment decision at every time t. With our representation, we can express the optimal cost between states and actions by the following relation:

$$\bar{C}_t(i) = \min_{j \geq i} \mathsf{E}\left[c(i,j) + \bar{C}_{t+1}(j - D_{t+1})\right], \tag{2.18}$$

The interpretation of this equation is that the optimal cost $\bar{C}_t(i)$ at time t when the last inventory $I_t = i$ is the expected cost of that state i and the optimal action j, for that period in the function $c(i,j)$. Plus, the expected optimal cost of the new inventory at a level $j - D_{t+1}$. The minimum denotes choosing the action $j \geq i$ that minimizes this cost.

With this recursion, it is possible to solve for all $C_t(i)$ for all $i = -M, \ldots, S_{max}$ and times $t = 0, 1, \ldots, T$, starting from the end at time T. And populating entries backwards (it is easier to imagine a matrix $\bar{C}$ with entries $c_{i,t} = \bar{C}_t(i)$). This problem solving technique is known as *Dynamic Programming* and Equation (2.18) is a *Bellman Equation*[4]. Storing the optimal replenishment levels $j^*_{t,i} = \text{argmin}_{j \geq i}\mathsf{E}\left[c(i,j) + \bar{C}_{t+1}(j - D_{t+1})\right]$ is also a good idea, as they will be a *dictionary* to the optimal replenishment level when the inventory is i at time t. See Figure 2.5 to visualize the dynamic inventory policy.

The modeling aspect is captured in the cost function $c(i,j)$. The goal is to modify this function to correctly capture the underlying business problem that we want to solve. We illustrate this with the following example:

Example 2.3 (Oxygen Concentrator Distribution Center). An oxygen concentrator company that delivers oxygen concentrators (a device used for patients that need a constant flow of oxygen while they are at home) rents a warehouse to be a distribution center for the concentrators in a major city. They have a random seasonal daily demand for concentrators D_t,

[4] A equation representing a recursive relation of the optimal decision of a dynamic problem.

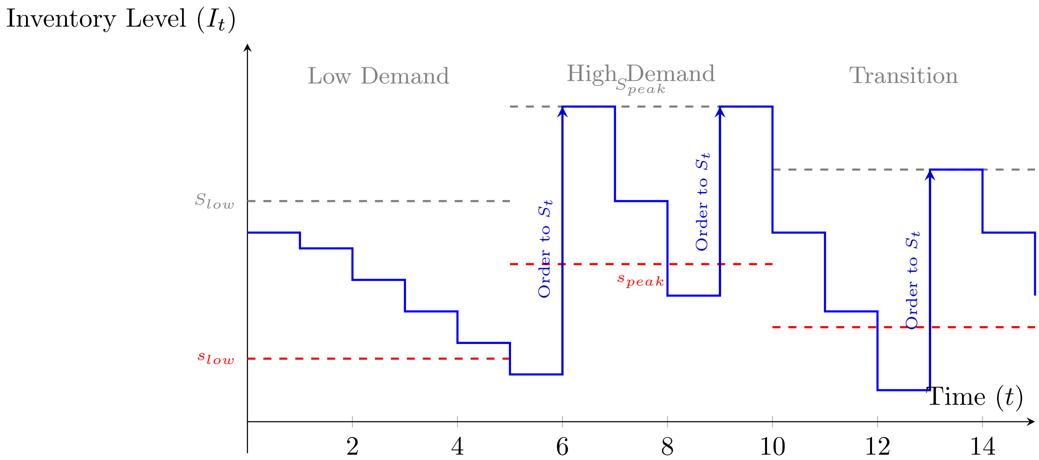


Figure 2.5: Inventory process with time-varying demand. As the depletion rate (slope) increases during peak periods, the Dynamic Programming solution adjusts the optimal s_t and S_t levels to minimize expected costs over the horizon T.

that among other features, depends on the atmospheric humidity level X_t (there is a correlation between respiratory diseases and the level of humidity in the air). The cost structure of running the warehouse works as follows: orders that are fulfilled with stock from the warehouse have a unit cost of c dollars (using an in-house delivery system). Orders that are fulfilled when the demand exceeds the stock cost $\tilde{c}$ dollars using an external courier that picks up a concentrator in another facility and delivers it to the client, this cost is higher than the in-house system, that is $\tilde{c} > c$. Stock replenishment orders have a fixed cost K per truck plus a unit cost c_r (the lead time is overnight, and each truck can store up to L concentrators). Holding the concentrators in the facility has an associated daily cost c_h (including maintenance and the internal logistics of the warehouse that are proportional to the number of concentrators). How to model and minimize this cost over the next month ($T = 30$, assuming today is $t = 0$) assuming a known high-fidelity forecast of the humidity levels X_t for next month $\{x_t\}_{t=0}^{T}$?

The first step is to write the cost function $c(i, j)$ to reflect these costs. For example, $c(i, j) = i^+ c_h + \tilde{c}(D_{t+1} - j)^+ + \mathbf{1}\{j > i\}(\lceil (j^+ - i^+)/L \rceil K + (j^+ - i^+)c_r)$, note the $i^+ = \max(i, 0)$ reflects that when the inventory ends up negative, these orders are not *back-ordered* as they still end up fulfilled at the higher cost $\tilde{c}$. $(j^+ - i^+)$ denotes the order quantity. The quantity $\lceil (j^+ - i^+)/L \rceil$ denotes the number of trucks needed ($\lceil a \rceil$ is the closest integer above a, typically referred in programing as the ceiling function[5]).

The second step is to incorporate the humidity forecast $\{x_t\}_{t=0}^{T}$. To do this, recall in Section 1 that a way to incorporate side-information is to analyze the conditional random variable, that

[5]A definition is $\lceil a \rceil = \min\{x \in \mathbb{Z} : x \geq a\}$.

is, $D_t|X_t$. Calibrating this conditional distribution results in conditional probabilities $\mathsf{P}(D_t = i|\{X_t = x_t\}_{t=0}^T)$. Solving the problem with this side information amounts to calculate the expectations with these conditional probabilities (instead of the unconditional ones). Replacing both the new cost $c(i,j)$ and taking expectations conditional to $X_t = x_t$ in (2.18). Explicitly, this yields the recursion for $\bar{C}_t(i) = \min_{j \geq i} \mathsf{E}[i^+ c_h + \tilde{c}(D_{t+1} - j)^+ + \mathbf{1}\{j > i\}(\lceil (j^+ - i^+)/L \rceil K + (j^+ - i^+)c_r) + \bar{C}_{t+1}(j - D_{t+1})|\{X_t = x_t\}_{t=0}^T]$.

Chapter 3

Network Fulfilment Models

3.1 Introduction

So far, we have described mathematical models to match the random demand in the time dimension. In this Chapter, we change the pace by thinking of problems satisfying the demand physically in the space dimension. Most if not all problems in supply chain analysis deal with some form of transportation problem as the demand requires moving goods or supplies between the producers and customers.

Matching the demand in space also adds a non-trivial time delay that needs to be accounted for when matching resources to satisfy the demand. Large companies like Amazon have to optimize the location of their warehouses and orchestrate a large chain of distribution networks to deliver their products in record time. Likewise, ridesharing marketplaces (such as Uber or Doordash) have to balance a demand that changes in space and time with a supply side of workers and drivers that will service different geographical locations as they finish their deliveries and rides.

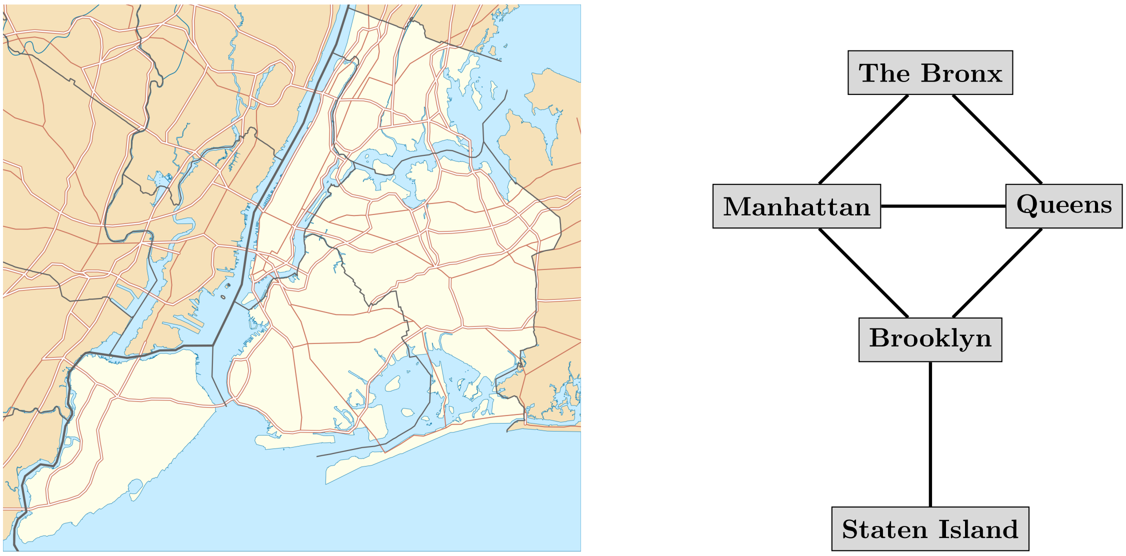
We will explore a series of common problems and solutions to supply chain management problems in space, as well as algorithms and heuristics to solve them.

3.2 Warehouse consolidation and positioning

Imagine a supplier wanting to build a warehouse to service demand in a city or geographical location. The supplier would want to build the warehouse in a place that is economically viable and resource efficient to service this demand.

While the physical world that we inhabit is 3-dimensional, it is often useful to think of abstractions that simplify the problem: For example, most geographical maps use a 2-dimensional representation of the Earth (this makes sense as the Earth is more or less a manifold locally, i.e. locally, we live in a flat world). Perhaps, one of the most powerful abstractions for physical spaces are graphs. A graph G is defined as a set of vertices V and edges E.

Graphs can be seen as discretization of a continuous space into a disjoint partition, connecting with edges all physically adjacent partitions. For example, take a map such as in the left side of Figure 3.1 with the five Boroughs of New York City (Manhattan, The Bronx, Queens, Brooklyn and Staten Island) and make each Borough a vertex, drawing an edge if there is a road connection between two Boroughs (vertices).



(a) Geographic Map of NYC's Five Boroughs.

(b) Graph abstraction of NYC.

Figure 3.1: Graph abstraction of NYC representing each of the five Boroughs as vertices, and their roads connections as edges.

Modeling a physical space as a graph has the natural trade-off of having tractable algorithms to solve a wide-range of problems while losing some of the accuracy and resolution of the underlying modeled real-world system. For example, in the graph of the figure, direct travels between Manhattan and Queens can occur via two different bridges, but there is only one edge representing their connection. In modeling traveling times, there is either the choice of creating more nodes and egdes to represent the different routes and to attach numerically the respective travel time to each edge, or to *average* into the single edge all the possible routes between Queens and Manhattan. Clearly, the tradeoff is between accuracy of the abstraction and tractability of the solution, as most optimization algorithms scale proportional to the size of the Graph. In general, mathematical modeling is the art of choosing an appropriate abstraction of a real-life system that preserves the behavior of interest while being algorithmically tractable.

Going back to the problem of choosing a location for a warehouse, start with a graph G, composed by N vertices $V = \{1, 2, \ldots, N\}$ and edges $E \subseteq \{(i, j)|(i, j) \in V^2\}$. For example, the graph in Figure 3.1 is composed by $N = 5$ vertices $V = \{1, 2, 3, 4, 5\}$, each vertex corresponds to a Borough: $1 \rightarrow$ Manhattan, $2 \rightarrow$ The Bronx, $3 \rightarrow$ Queens, $4 \rightarrow$ Brooklyn and $5 \rightarrowtail$ Staten Island. And edges $E = \{(1,2),(1,3),(1,4),(2,3),(3,4),(4,5)\}$.

Vertices can have different unique attributes, for example, in the model of this section each vertex has an associated random demand D_i for $i = 1,\ldots,N$. This demand, is the usual demand random variable and denotes the random realization of the demand (in this case, there is not yet a temporal element to the demand and it can be taken as a static model with i.i.d. realizations of the demand that repeat every day). Likewise, every edge (i,j) for $i,j \in \{1,\ldots,N\}$ also has a random variable T_{ij} denoting the random traveling time of a vehicle starting in vertex i to vertex j.

3.2.1 Single Location Warehouse

With these elements, we can now start to model the problem of choosing a location to build a warehouse. Let c_i be the cost of running a warehouse in location i with enough capacity to service the demand $\sum_{i=1}^N D_i$ with high probability (see Chapters 1 and 2 on how this number can be calculated, here it is taken as given[1]). In this scenario, if it's not possible to run a warehouse in some vertex, then let the cost be $c_i = \infty$ for such vertex (or exclude it from the set of possible locations to run the warehouse). Likewise, let the cost of transports between vertices (i,j) be equal to[2] $c_{ij} \propto T_{ij}$, for example the cost of the labor of a courier plus the cost of gas that is proportional to the random traveling time T_{ij}, that is $c_{ij} = \gamma_{ij} T_{ij}$, clearly c_{ij} is also a random variable.

For a warehouse built on location i, the cost is equal to $c_i + \sum_{j=1}^N D_j c_{ij}$. As this number is a random variable, a first approximation is to look at the expected value and choose over all possible locations, that is, solving the enumeration problem:

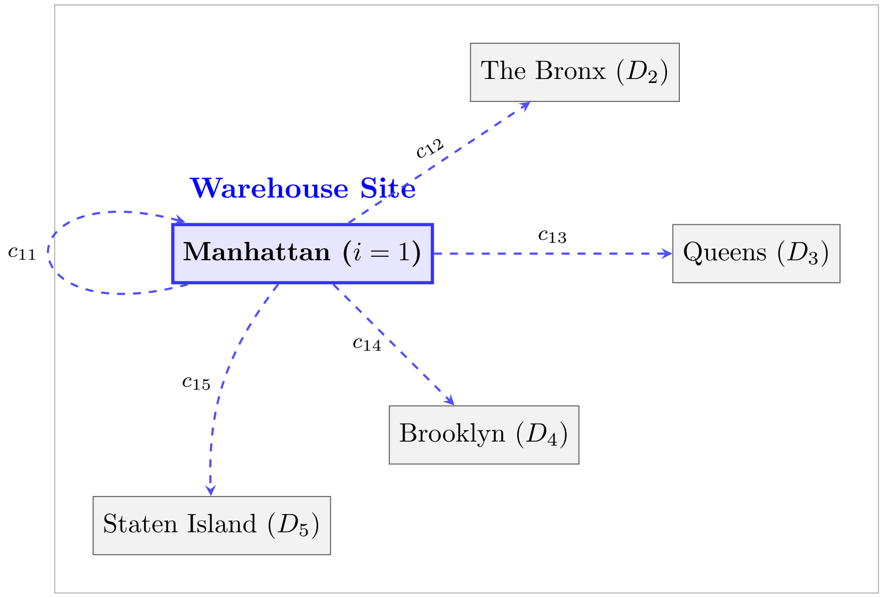
$$\min_{i=1,\ldots,N} \mathsf{E}\left[c_i + \sum_{j=1}^{N} D_j c_{ij}\right], \tag{3.1}$$

It is an enumeration problem because it amounts to calculate the cost for each location and choose the minimum. The emphasis in writing the costs as random variables instead of the more common practice of writing them as constants is that there is a conscious choice to solve the problem only in expected value or to consider other moments of their distribution. For example, it could be the case that running the warehouse in one location has the lowest cost in expected value, but the variation (either of the demands and the trips) is higher than another location that has a slightly higher cost in expected value, but less variance (in this case, variance denotes

[1] In practice, one way would be to calculate a capacity ϵ such that $\mathsf{P}(\sum_{i=1}^N D_i > \epsilon) \leq 1-\alpha$ and then quote or calculate the actual cost of running a warehouse of that capacity for the value of c_i. You can also set up a Newsvendor-like model to calculate a capacity that is also sensitive to missed sales or holding costs.

[2] $a \propto b$ denotes that b is proportional to a up to a constant, say $b = 2a$.

the daily variation of running the warehouse satisfying random demands every day or period). See Figure 3.2 to get a feel for evaluating the cost of building a single warehouse in Manhattan.



Evaluating Total Cost for $i = 1$: $c_1 + \sum_j D_j c_{1j}$

Figure 3.2: Visualization of the enumeration step for $i = 1$ (Manhattan). To solve the problem, the supplier calculates the expected total cost (warehouse overhead c_i plus weighted transportation costs to all demand nodes $D_j c_{ij}$) for each potential vertex and selects the minimum.

In general, for problems involving a random amount of money (either a cost or a payoff) there is an amount of risk aversion that the decision-maker is willing to take, and it needs to be made explicit in the problem formulation, or otherwise be acknowledged by the modeler. For example, given two lotteries: one that pays $X_1 = \$100,000 B_1 + (-\$100,000)(1 - B_1)$ and another $X_2 = \$100 B_2 + (-\$100)(1 - B_2)$ with two independent coins B_1, B_2 with $\mathsf{P}(B = 1) = \mathsf{P}(B = 0) = 1/2$. That is, two loteries that with 50% chance pay a positive amount and with the same chance the player loses the same amount, in the first case \$100,000 and in the other \$100. While both lotteries pay in average \$0, that is $\mathsf{E}(X_1) = \mathsf{E}(X_2) = \$0$ they are clearly not the same game. Depending on their risk aversion, some players would choose to play lottery 1, while others would play lottery 2 (or even choose not to play at all). The quantitative difference between the two games is their variance. We have that the variance of the first lottery is $\mathsf{Var}(X_1) = 2^2(\$100,000)^2 \mathsf{Var}(B_1) = (\$100,000)^2$ and the second one $\mathsf{Var}(X_2) = (\$100)^2$. The risk ratio between the two lotteries is $\mathsf{Var}(X_1)/\mathsf{Var}(X_2) = 10^6$. In other words, playing the first lottery has a million times higher variance than the second. While exaggerated, this example shows the potential literal risk of only modeling decisions based on their expected value. A quantitative approach to evaluate each lottery could be to *weight* the variance or the standard deviation, for example $\mathsf{E}(X_1) + \lambda\sqrt{\mathsf{Var}(X_1)}$ weights the standard deviation (in dollars). A weight $\lambda = 0$ denotes risk indifference (thus, only caring for the expected return of the lottery). $\lambda < 0$ denotes risk aversion, while $\lambda > 0$ implies risk appetite. Note that the units of $\mathsf{E}(X_1) + \lambda\sqrt{\mathsf{Var}(X_1)}$ is dollars.

Another justification coming from the study of risk measures also justifies this by defining the risk measure of a random variable X as $R_\lambda(X) = \frac{1}{\lambda} \ln \mathsf{E}(e^{\lambda X})$. This risk measure is used in risk management as it captures risk aversion, is convex and coherent and it is sensitive to tail risks. In fact, we show that our strategy of weighting the risk is approximately equivalent to this risk measure. To see why, note that $e^u = 1 + u + u^2/2 + \dots$. Then, $\mathsf{E}(e^{\lambda X}) = \mathsf{E}(1 + \lambda X + \frac{\lambda^2 X^2}{2} + \dots) = 1 + \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{E}(X^2) + \dots = 1 + v$ where $v = \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{E}(X^2) + \dots$, also note that $\ln(1+v) = v - \frac{v^2}{2} + \dots$. Using these relations and noting that terms with v^3 and higher vanish, we get:

$$\begin{aligned}
\ln \mathsf{E}(e^{\lambda X}) &= \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{E}(X^2) - \frac{1}{2}\left[\lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{E}(X^2)\right]^2 + \dots \qquad (3.2)\\
&= \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{E}(X^2) - \frac{\lambda^2}{2}[\mathsf{E}(X)]^2 + \dots\\
&= \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}[\mathsf{E}(X^2) - \mathsf{E}(X)^2] + \dots\\
&= \lambda \mathsf{E}(X) + \frac{\lambda^2}{2}\mathsf{Var}(X) + O(\lambda^3)
\end{aligned}$$

Noting that the term $O(\lambda^3)$ vanishes to 0 and dividing both sides by λ yields the desired relation.

This shows that optimizing the objective $\mathsf{E}(X) + \lambda \mathsf{Var}(X)$ is a principled way of handling the risk. An important aspect in practice is calibrating the risk aversion parameter λ. A well-known principle to do this, is known as the *Certainty Equivalent*. This is simply an amount of money that the decision maker is willing to pay (or receive) to be indiferent between the lottery and a certain payoff. Solving for that amount would yield the implied risk aversion coefficient. For example, suppose there is a lottery $X = -\$100B + \$0(1-B) = -\$100B$, that is, losing a \$100 with probability $\mathsf{P}(B=1)$ or losing 0 with probability $\mathsf{P}(B=0)$. Suppose $P(B=1) = 1/2$, we have $\mathsf{E}(X) = -\$50$ and $\mathsf{Var}(X) = (\$100)^2/4$. A person is willing to pay \$40 to exit the lottery, this implies a Certainty Equivalent (CE) given by $-40 = -50 + \frac{\lambda}{8}100^2$ or $\lambda = 80/100^2$. In general, giving the stakeholders similar lotteries and with their CEs is a standard way to learn the risk aversion of a stakeholder, we will illustrate numerically how the trade-off between risk and expected cost plays out later.

Going back to the warehouse location problem, the problem incorporating variance can be stated as follows:

$$\min_{i=1,\dots,N} \mathsf{E}\left[c_i + \sum_{j=1}^{N} D_j c_{ij}\right] + \lambda\sqrt{\mathsf{Var}\left(\sum_{j=1}^{N} D_j c_{ij}\right)}, \qquad (3.3)$$

With $\lambda > 0$ this optimization problem models a risk-averse decision-maker that penalizes variance in their cost (caused by both variance in the demand and in the transportation times).

This is again an enumeration problem that amounts to calculate the objective for each potential warehouse location. It is also possible to consider other risk measures that penalize variation.

In the case where the cost of the facilities c_i is fixed. The variance of the cost $\mathsf{Var}[c_k + \sum_{j=1}^{N} D_j c_{kj}]$ can be calculated as $\mathsf{Var}[\sum_{j=1}^{N} D_j c_{kj}] = \mathbf{1}\Sigma_k \mathbf{1}^\intercal$ where the entry i, j of the matrix Σ_k is $\mathsf{Cov}(D_i c_{ki}, D_j c_{kj}) = \gamma_{ki}\gamma_{kj}\mathsf{Cov}(D_i T_{ki}, D_j T_{kj})$. For example, in the case of the diagonal we have $\mathsf{Cov}(D_i c_{ki}, D_i c_{ki}) = \gamma_{ki}^2 \mathsf{Var}(D_i T_{ki})$ if the demand and the travel times are independent then this is equal to $\gamma_{ki}^2 \mathsf{Var}(D_i)\mathsf{Var}(T_{ki})$. For the other elements off-diagonal we use the definition of covariance[3], then, $\mathsf{Cov}(D_i T_{ki}, D_j T_{kj}) = \mathsf{E}(D_i T_{ki} D_j T_{kj}) - \mathsf{E}(D_i T_{ki})\mathsf{E}(D_j T_{kj})$. With the assumption that while the demands and the travel times are independent, but not the demands and the travel times with each other. For example, it makes sense that travel jams in Manhattan could affect traffic in The Bronx, and also that the demand for a product is Brooklyn is correlated with the demand for the same product in Queens. With this assumption, we have that $\mathsf{E}(D_i T_{ki} D_j T_{kj})$ is equal to $\mathsf{E}(D_i D_j)\mathsf{E}(T_{ki} T_{kj})$ and $\mathsf{E}(D_i T_{ki})\mathsf{E}(D_j T_{kj})$ is equal to $\mathsf{E}(D_i)\mathsf{E}(T_{ki})\mathsf{E}(D_j)\mathsf{E}(T_{kj})$. In summary, the covariance matrix Σ_k has entries i, j equal to:

$$\Sigma_k(i,j) = \begin{cases} \gamma_{ki}^2 \mathsf{Var}(D_i)\mathsf{Var}(T_{ki}), & \text{for } i = j \\ \gamma_{ki}\gamma_{kj}[\mathsf{E}(D_i D_j)\mathsf{E}(T_{ki}T_{kj}) - \mathsf{E}(D_i)\mathsf{E}(D_j)\mathsf{E}(T_{ki})\mathsf{E}(T_{kj})], & \text{for } i \neq j. \end{cases} \tag{3.4}$$

3.2.2 Multiple Location Warehouses

Sometimes it can be more efficient to choose multiple locations to satisfy the geographical demand of multiple locations than a single warehouse location. This can be because the cost structure is more favorable (closer deliveries are cheaper), due to economies of scale, or simply ease of logistical burden by breaking down the demand to multiple difference warehouses. The problem can also be interpreted as choosing providers to satisfy demand for different geographical locations.

To modify the previous section formulation, now each candidate location has a finite capacity C_i to service demand. Likewise, we modify the cost c_i to be a fixed (and given) cost of running the warehouse i. The second and more important modification is that since multiple locations can be chosen to service the demand, brute enumeration is in general not a great strategy, except for small instances of the problem. To see why, suppose there are M candidate locations, all possible choices of subsets of locations is 2^M, which is a number that grows exponentially in the number of possible locations[4]. Combinatorial Optimization studies algorithms to solve problems where the solution takes the form of subsets by exploiting the internal structure of the problem, not to be forced to enumerate all possible 2^M candidate solutions. A very powerful technique to solve this problems is called Mixed Integer Programming, that is very similar to Linear Programming (both in structure and algorithmically) by allowing integer variables (normally

[3] $\mathsf{Cov}(X,Y) = \mathsf{E}(XY) - \mathsf{E}(X)\mathsf{E}(Y)$.

[4] Given a set of M elements, each can be either included or not included in a subset. Since this is true for all elements, the total number of subsets is 2^M, also called the *power set*.

encoding either counting quantities or codyfing yes-no decisions as variables). See Figure 3.3 for a typical representation of the problem of matching possible M warehouses locations with N demand target locations.

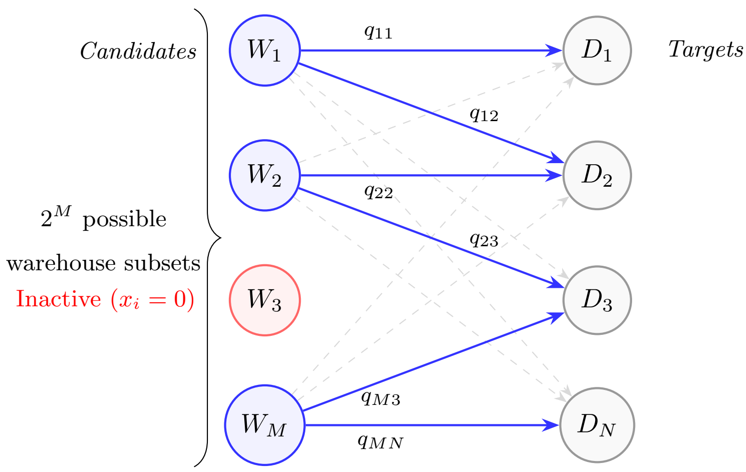


Figure 3.3: Bipartite representation of the Multiple Location Warehouse problem. Solid blue lines represent an active assignment q_{ij} from selected warehouses ($x_i = 1$). The light dashed lines represent the combinatorial complexity: the model must select a subset of M locations and then optimize the flows to N targets.

A Mixed Integer Programming formulation of this problem entails adding binary variables $x_i \in \{0,1\}$ denoting whether or not the location is chosen. Also, the variables $q_{ij} \geq 0$ denote the amount of demand (quantity) satisfied from warehouse i to location $j = 1, \ldots, N$. The next modeling step is to relate variables x_i and q_{ij} to each other in an equation (that normally becomes a constraint of the problem). For each warehouse it must be true that when it is selected ($x_i = 1$), the total quantity that can be serviced from the location cannot exceed its capacity C_i. In other words, $\sum_{j=1}^N q_{ij} \leq C_i$ when $x_i = 1$, and conversely when $x_i = 0$ it should be $\sum_{j=1}^N q_{ij} = 0$. MIP formulations normally involve tricks manipulating real-world conditions into equations (see [1] for some of the most common tricks). In this case, the trick is to write the constraint as $\sum_{j=1}^N q_{ij} \leq x_i C_i$. It is immediate that this constraint achieves the goal of encoding both situations.

Likewise, it must be true that the random demand is satisfied to some level. For example, if looking at the problem in expected value, that the demand is satisfied in average is the constraint $\sum_{i=1}^M q_{ij} \geq \mathsf{E}(D_j)$. Another common technique to deal with this type of constraints is to write a constraint as $\mathsf{P}(\sum_{i=1}^M q_{ij} \leq D_j) \leq \eta$, meaning that with probability not exceeding η (say a small percentage like 5%) the random demand will be satistied. In the case where the demand can be approximated as a normal random variable, that is, $D_j \overset{\mathrm{d}}{=} \mathsf{E}(D_j) + \sqrt{\mathsf{Var}(D_j)}Z$. The constraint $\mathsf{P}(\sum_{i=1}^M q_{ij} \leq D_j) \leq \eta$ can then be written as $\mathsf{P}(Z \leq [\sum_{i=1}^M q_{ij} - \mathsf{E}(D_j)]/\sqrt{\mathsf{Var}(D_j)}) \geq 1-\eta$, or the constraint $\sum_{i=1}^M q_{ij} \geq \mathsf{E}(D_j) + \sqrt{\mathsf{Var}(D_j)}\Phi^{-1}(1-\eta)$, the similarity to the Newsvendor model studied in Chapter 2 should be evident, as the contraint conveys that the quantity delivered should be the mean of the demand plus a buffer proportional to the standard deviation of the demand. The last constraint to be enforced is that $q_{ij} = 0$ when $x_i = 0$ and positive otherwise. Letting $\bar{d}_j$ be an upper bound for the demand such that $\mathsf{P}(D_j > \bar{d}_j) = 0$, this can be achieved with the constraint $q_{ij} \leq x_i \bar{d}_j$ With these constraints, we can write a first fomulation of the problem in expected value as (let $W \subseteq V$ be the set of possible locations for the warehouses to make the notation cleaner):

$$\begin{aligned} \min_{q_{ij} \geq 0, x_i \in \{0,1\}} \quad & \mathsf{E}\left\{ \sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right\} \\ \text{s.t.} \quad & \sum_{i \in W} q_{ij} \geq \mathsf{E}(D_j) \ \text{ for } \ j \in V, \\ & \sum_{j \in V} q_{ij} \leq x_i C_i \ \text{ for } \ i \in W, \\ & q_{ij} \leq x_i \bar{d}_j \ \text{ for } \ i, j \in W \times V. \end{aligned} \tag{3.5}$$

As the costs c_i are fixed, the objective can be written as $\sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} \mathsf{E}(c_{ij}) q_{ij}$. This formulation is the typical warehouse optimization found in textbooks and can be solved with off-the-shelf MIP solvers.

Similar to the discussion of the single warehouse model, as is, the model is only minimizing the cost in expected value (both in the objectuve and the first constraint). Variations in the travel times T_{ij} will cause fluctuations in the cost. We can take a similar strategy to hedge this variation using a similar reasoning as in the single warehouse case. To do so, note that the variance $\mathsf{Var}[\sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij}]$ is equal to $\sum_{i \in W} \sum_{j \in V} \sum_{k \in W} \sum_{l \in V} q_{ij} \mathsf{Cov}(c_{ij}, c_{lk}) q_{lk}$ which can be written in matrix form as $\boldsymbol{q} \Sigma \boldsymbol{q}^\intercal$ where Σ has entries:

$$\mathsf{Cov}(c_{ij}, c_{lk}) = \begin{cases} \gamma_{ij}^2 \mathsf{Var}(T_{ij}), & \text{for } i, j = k, l \\ \gamma_{ij} \gamma_{kl} [\mathsf{E}(T_{ij} T_{kl}) - \mathsf{E}(T_{ij}) \mathsf{E}(T_{kl})], & \text{for } i, j \neq k, l. \end{cases} \tag{3.6}$$

Also using the chance constraint for the demand, the risk aware formulation of the problem can be written as a Mixed Integer Quadratic Program (MIQP):

$$\begin{aligned} \min_{q_{ij} \geq 0, x_i \in \{0,1\}} \quad & \mathsf{E}\left\{ \sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right\} + \lambda \mathsf{Var}\left(\sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right) \\ \text{s.t.} \quad & \sum_{i \in W} q_{ij} \geq \mathsf{E}(D_j) + \sqrt{\mathsf{Var}(D_j)} \Phi^{-1}(1 - \eta) \ \text{ for } \ j \in V, \\ & \sum_{j \in V} q_{ij} \leq x_i C_i \ \text{ for } \ i \in W, \\ & q_{ij} \leq x_i \bar{d}_j \ \text{ for } \ i, j \in W \times V. \end{aligned} \tag{3.7}$$

This formulation of the problem can be solved with a MIQP solver. Note that in this case the objective uses the variance instead of the standard deviation as in the single warehouse section. The problem can still be solved as a Mixed Integer Second-Order Conic Program (MISOCP). The formulation is also implicitly assuming that the travel times and the demands are independent from each other. In Figure 3.4 we can see the trade-off we mentioned earlier considering different values of λ.

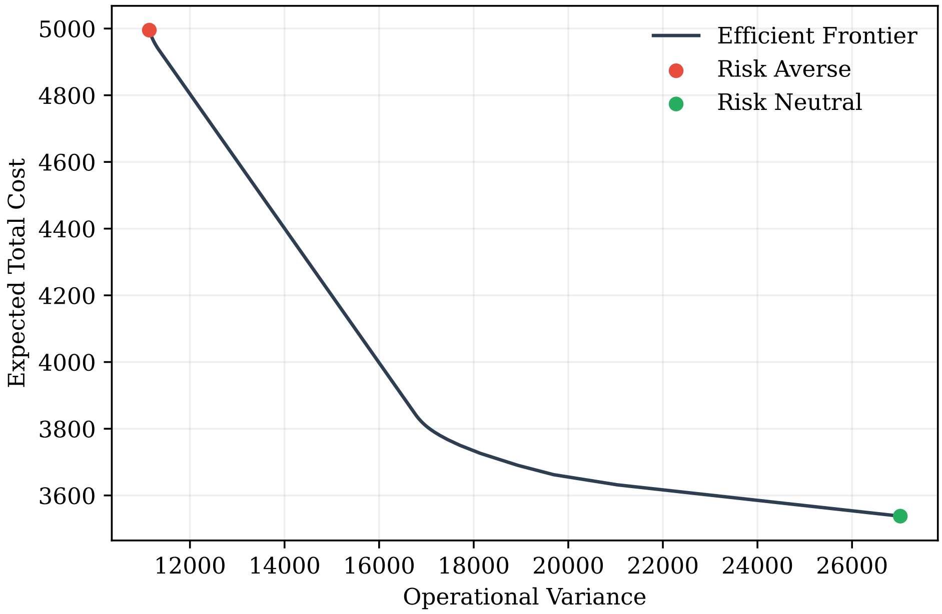


Figure 3.4: The Efficient Frontier for the warehouse location problem. The curve illustrates the trade-off between minimizing expected operational costs and reducing cost variance (risk). The green marker represents a risk-neutral approach ($\lambda = 0$), while the red marker indicates increasing risk aversion.

3.3 Distributionally Robust Warehouse Location Optimization

Readers of the first sections should be feeling slightly uneasy regarding the sensitivity of these optimization problems. By sensitivity, we mean how the optimal solution changes when the input data changes. For example, there is a lot of expected values for the demand $\mathsf{E}(D_j)$, that as we have seen, when there is a small number of observations or it is in reality time-varying, the sample mean $\sum_{t=1}^{T} d_t$ might be within ax range of the true value. Compounding this effect across all expected values and variances (which are in turn expected values as well) creates an effect where the optimal solution of solving the warehouse problem with a sample observations $\hat{q}^*_{ij}$ might be substantially different from the true optimal solution q^*_{ij}.

Given a fixed solution q^*_{ij} we have explored previously how to build confidence intervals using the CLT (see Chapter 2 and the empirical distributions section to see some of these techniques). To get a feel for the interplay between the sample size n and the number of vertices N we use a different but useful way of building confidence intervals, we illustrate this with an example:

Example 3.1 (Local fulfillment centers). A last-mile delivery company wants to know the optimal location to build their fulfillment centers to serve the New York metropolitan area. Geographically, this area is composed by 23 counties, which makes a graph with $N = 23$ nodes. The company has a sample size of daily orders for each of the counties going over 4 years for a total of around $n = 1,460$ observations. Their data-scientist solves the basic warehouse location problem in Equation (3.5) obtaining a solution x_i^*, q_{ij}^* for the optimal locations of the warehouses and the quantity that each should deliver to each county. The management of the company seems skeptical of the solution and ask the data-scientist for confidence intervals for their estimation. Specifically, they want to know with $1 - \alpha = 95\%$ confidence what is a range for the variable cost component of the cost (involving the quantities q_{ij}), as they know with certainty how much it costs to run each of the facilities once active.

To solve this problem, we use a very useful result known as *Hoeffding's Inequality* which states the following for a sum of independent random variables $S_n = X_1 + \cdots + X_n$, where all variables are bounded, that is, $\mathsf{P}(X_i \in [\ell, u]) = 1$. Then, we have[5]

$$\mathsf{P}\left(S_n - \mathsf{E}[S_n] \geq \varepsilon\right) \leq \exp\left(\frac{-2\varepsilon^2}{n(u-\ell)^2}\right), \tag{3.8}$$

The trick to use this bound is to express the objective value as a function operating on a real valued function. We initially have a random vector for the demands $\boldsymbol{D} = (D_1, \ldots, D_N)$, given a solution $x = (x_1, \ldots, x_M)$ and $\boldsymbol{q} = (q_{11}, \ldots, q_{MN})$ with cost vector $c = (c_{11}, \ldots, c_{MN})$. The value that we want to bound is $\mathsf{E}[c\boldsymbol{q}^\mathsf{T}]$. The first observation is noting that for any solution, in practice is should be the case that $\sum_{i \in W} q_{ij} = D_j$ for all nodes $j \in V$. Then, by letting the vector $\bar{c} = (\bar{c}_1, \ldots, \bar{c}_N)$ where $\bar{c}_j = (\sum_{i \in W} c_{ij} q_{ij})/(\sum_{i \in W} q_{ij})$ for $j \in V$. In other words, the vector $\bar{c}_j$ is the average cost to deliver to each location. With this, we have $c\boldsymbol{q}^\mathsf{T} = \bar{c}\boldsymbol{D}^\mathsf{T}$. Now we can use Hoeffding's inequality to build our cost bound. The function $\bar{c}\boldsymbol{D}^\mathsf{T}$ can be seen as the generic random variable X in the inequality. The first step is to determine its bound. Note that all demands are bounded below by 0, that is, $\ell = 0$. Likewise, each demand is bounded by above by $\bar{d}_j$, then, all of them are bounded by $\bar{d} = \max_{j \in V} \bar{d}_j$. Then, it follows that the cost is bounded as $\bar{c}\boldsymbol{D}^\mathsf{T} \leq |\bar{c}|\bar{d}$ (we abuse notation by letting $|\cdot|$ be the $L-1$ norm for vectors. For example, $|\bar{c}| = \sum_{j \in V} |\bar{c}_j|$). This norm $|\bar{c}|$ is bounded by $N \max\{c_{ij}\}$ where $\max\{c_{ij}\}$ is the highest possible cost, the N is simply the number of vertices in the graph. Then, the random variable $\mathsf{P}(\bar{c}\boldsymbol{D}^\mathsf{T} \in [0, N\max\{c_{ij}\}\bar{d}]) = 1$.

An i.i.d sample of the demand vector $\boldsymbol{D}$ is represented as vectors $\boldsymbol{d}_k$ for $k = 1, \ldots, n$, where $\boldsymbol{d}_k$ is a vector with N elements, the demand for each region. The sample expected cost is

[5] The more general version, states that when each random variable is bounded as $\mathsf{P}(X_i \in [\ell_i, u_i]) = 1$ the denominator of the exponential is $\sum_{i=1}^n (u_i - \ell_i)^2$ instead of $n(u-\ell)^2$.

then $\mathsf{E}_{\mathsf{P}_\mathsf{n}}[\bar{c}\boldsymbol{D}] = \frac{1}{n}\sum_{k=1}^{n} \bar{c}\boldsymbol{d}_k$ while the true cost is the expected value $\mathsf{E}[\bar{c}\boldsymbol{D}]$. Using Hoeffding's inequality we get that:

$$\mathsf{P}\left(\mathsf{E}_{\mathsf{P}_\mathsf{n}}[\bar{c}\boldsymbol{D}] - \mathsf{E}[\bar{c}\boldsymbol{D}] \geq \epsilon\right) \leq \exp\left(\frac{-2n^2\varepsilon^2}{n(N\max\{c_{ij}\}\bar{d})^2}\right), \tag{3.9}$$

With $1-\alpha = 95\%$ probability, we get that the true cost is bounded by (solving for ϵ):

$$\mathsf{E}[\bar{c}\boldsymbol{D}] \leq \mathsf{E}_{\mathsf{P}_\mathsf{n}}[\bar{c}\boldsymbol{D}] + N\max\{c_{ij}\}\bar{d}\sqrt{\frac{1}{2n}\ln\left(\frac{1}{\alpha}\right)}. \tag{3.10}$$

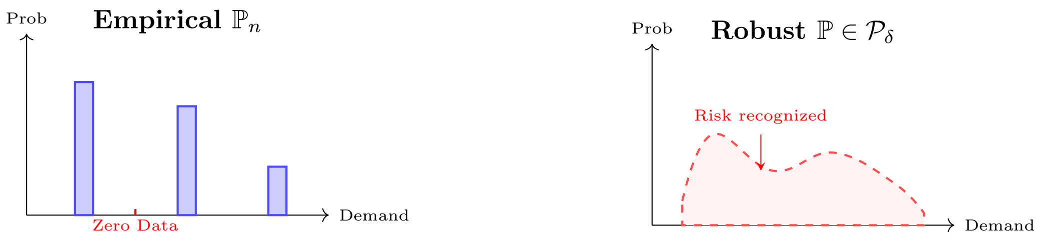
This means that solving the problem with a sample of n observations is likely underestimating the true expected cost due to the uncertainty caused by the number of geographical locations N. To mitigate this uncertainty requires at least a quadratic increase in the sample size n. In the next section, we present a strategy to deal with solving an alternative optimization problem that deals with the fact that the true distribution is in a neighborhood of the true distribution.

3.3.1 A reformulation using duality and optimal transport

Back in the first chapter, we defined both the actual expected value of a random variable and their sample counterpart. A specially important random variable is the indicator function $\mathbf{1}_e$ that is equal to 1 if event e occurs, and 0 otherwise. e could be for example the event that today rains, or that the demand exceeds a thousand units. The expected value of an indicator $\mathsf{E}(\mathbf{1}_e)$ is equal to the probability of the event $\mathsf{P}(e)$. And the sample estimate is the same as we defined before, that is, $p_n(e) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{e_i\}$. For example, the sample probability that the demand is equal to 10, is the sample probability $p_n(D = 10) = \sum_{i=1}^{n} \mathbf{1}\{d_i = 10\}/n$. Sample probabilities behave the same as expected values, that is, they converge to their true value if the distribution is i.i.d.

For a random variable X taking values in the set $\mathcal{X}$ (for example, in the case of the demand random variable D, the demand takes values in the set of the natural numbers $\mathbb{N}$), the collection of sample probabilities $p_n(X = x) = \sum_{i=1}^{n} \mathbf{1}\{X_i = x\}$ for all $x \in \mathcal{X}$, forms an *empirical distribution* of X. This distribution is typically denoted by P_n instead of the true probability distribution P. The reader can already imagine that as the sample size n increases, the empirical distribution will converge to the true one by the LLN, that is $\mathsf{P}_n \to \mathsf{P}$. It is also, expected that the empirical distribution of n observations will overestimate the probability of some events, while also underestimating others. In fact, since both probability distributions add up to 1, the amount of overestimation is exactly the amount of underestimation (think of why this is the case).

Given these facts, the methodology known as *Distributionally Robust Optimization* (DRO) aims to incorporate the idea that the true distribution P of the data is within some neighborhood of the empirical distribution P_n. Here, neighborhood is taken as a *distance metric* between the true and empirical distribution. Let $\mathcal{D}$ denote a metric distance function between distributions[6]. Then, we want to consider distributions P that are around the observed empirical distribution P_n within a distance δ, that is, distributions $\mathsf{P} : \mathcal{D}(\mathsf{P}_n, \mathsf{P}) \leq \delta$. In plain terms, a decision-maker acknowledges that their finite data samples might bias their estimates and wants to optimize considering the case where the true distribution is in some neighborhood of the empirical distribution P_n, in this case, neighborhood means the set of possible distributions that are in a *ball* of radius δ around P_n. The decision-maker picks the radius according the their risk aversion (similar to choosing λ in the previous examples). A larger radius δ means less trust in the empirical distribution (for example, due to a low sample size) while a small radius means that there is high confidence in the empirical distribution. See Figure 3.5 for a conceptual representation of the DRO approach.



(a) Sparse & Fragile

(b) Spread & Robust

Figure 3.5: Comparison of fulfillment modeling. The empirical approach (left) ignores events not present in the sample, leading to $q_{ij} = 0$ for potentially critical regions. The DRO approach (right) considers a distribution that spreads probability mass into those gaps, forcing the optimization to allocate resources ($q_{ij} > 0$) for unobserved but plausible demand.

The DRO formulation can be set up in many ways. We will focus in a paradigm that uses duality theory and optimal transport. To simplify the problem we will only consider variation in the demand, that is, the costs c_{ij} are assumed to be known and fixed. Moreover, the historical (sample) demand is the matrix $D_n = (\boldsymbol{d}_k)$ for $k = 1, \ldots, n$, where $\boldsymbol{d}_k$ is a vector with the historical demands for each $i \in V$, that is, $\boldsymbol{d}_k$ is a vector of size N. Each of these vectors is an i.i.d. sample of the vector $\boldsymbol{D}$. Denote the set of distributions within δ distance of the empirical distributions as $\mathcal{P}_\delta = \{\mathsf{P} : \mathcal{D}(\mathsf{P}_n, \mathsf{P}) \leq \delta\}$. Let $x = (x_1, \ldots, x_M)$. First, we re-write the warehouse problem in this setting as:

$$\min \quad \left\{ \tilde{c}x^\intercal + \sup_{\mathcal{P}_\delta} \mathsf{E}\left[\min \boldsymbol{c}\boldsymbol{q}^\intercal | A\boldsymbol{q}^\intercal + Bx^\intercal \geq (\boldsymbol{D}, \boldsymbol{0})^\intercal\right] \right\}, \tag{3.11}$$

This formulation is a matrix form re-write of the problem of interest that is going to be useful for some algebraic manipulations that we are performing. For example, $\tilde{c}x^\intercal$ is equal to $\sum_{i \in W} c_i x_i$

[6]A distance metric is a function $\mathcal{D} : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ that operates on a set $\mathcal{X}$ with the properties: For $p_1, p_2, p_3 \in \mathcal{M}$ it is satisfied $\mathcal{D}(p_1, p_1) = 0$, $\mathcal{D}(p_1, p_2) > 0$ for $p_1 \neq p_2$, $\mathcal{D}(p_1, p_2) = \mathcal{D}(p_2, p_1)$ for all $p_1, p_2 \in \mathcal{M}$ and $\mathcal{D}(p_1, p_3) \leq \mathcal{D}(p_1, p_2) + \mathcal{D}(p_2, p_3)$ which is the so-called triangle inequality.

and $c\boldsymbol{q}^\intercal = \sum_{i\in W}\sum_{j\in V} c_{ij}q_{ij}$. The constraints $A\boldsymbol{q}^\intercal + Bx^\intercal \geq (\boldsymbol{D}, \boldsymbol{0})^\intercal$ are the usual constraints of the problem, such as demand satisfaction, e.g. see the constraints in (3.5), A would be the matrix of coefficients multiplying the $\boldsymbol{q}$ variables, while the right hand side is the random demand $\boldsymbol{D}$ for the demand satisfaction constraints and $\boldsymbol{0}$ for the constraints that only depend on variables. The interpretation of the problem is quite similar: minimize the overall cost of building the facilities, subject to the worst distribution $\mathcal{P}_\delta$. Typically, the intuitive explanation is that nature chooses the worst distribution given x (the sup part) and then the optimization chooses the optimal quantities $\boldsymbol{q}$.

The first step to solve this problem is to re-formulate the sup by exploiting duality and results from Transport Theory. Letting $Q(x,\boldsymbol{D}) = \min c\boldsymbol{q}^\intercal | A\boldsymbol{q}^\intercal + Bx^\intercal \geq (\boldsymbol{D}, \boldsymbol{0})^\intercal$, we are going to write the Lagrangian of $\sup_{\mathcal{P}_\delta} \mathsf{E}[Q(x,\boldsymbol{D})]$ (noting that the definition of $\mathcal{P}_\delta$ is a constraint defined by the distance of the distributions), for a pair of variables P, λ:

$$\sup_{\mathcal{P}_\delta} \mathsf{E}[Q(x,\boldsymbol{D})] = \min_{\lambda\geq 0}\sup_{\mathsf{P}}\{\mathsf{E}_{\mathsf{P}}[Q(x,\boldsymbol{D})] + \lambda(\delta - \mathcal{D}(\mathsf{P},\mathsf{P}_n))\}, \tag{3.12}$$

The next step is to choose a distance metric $\mathcal{D}$ that makes the problem tractable. One such metric, is the so-called *Wasserstein-1* distance. Choosing this distance $\mathcal{D} = W_1$ has the following property for any function $g(\boldsymbol{D})$:

$$\sup_{\mathsf{P}}\left[\mathsf{E}_{\mathsf{P}}[g(\boldsymbol{D})] - \lambda W_1(\mathsf{P},\mathsf{P}_n))\right] = \mathsf{E}_{\mathsf{P_n}}\left[\sup_{\boldsymbol{D}}\{g(\boldsymbol{D}) - \lambda|\boldsymbol{D} - \boldsymbol{D}_k|\}\right], \tag{3.13}$$

Where D_k is a generic sample of the empirical distribution P_n and $|\cdot|$ for vectors denotes the $L-1$ norm. We can let $g(\boldsymbol{D}) = Q(x,\boldsymbol{D})$. The math behind the result is somewhat technical and outside the scope of this section. By letting $g(\boldsymbol{D}) = Q(x,\boldsymbol{D})$ and rearranging $\min_{\lambda\geq 0}\sup_{\mathsf{P}}\{\mathsf{E}_{\mathsf{P}}[Q(x,\boldsymbol{D})] + \lambda(\delta - \mathcal{D}(\mathsf{P},\mathsf{P}_n))\}$ as $\min_{\lambda\geq 0}(\lambda\delta + \sup_{\mathsf{P}}\{\mathsf{E}_{\mathsf{P}}[Q(x,\boldsymbol{D})] - \lambda\mathcal{D}(\mathsf{P},\mathsf{P}_n)\})$ combining the two equations yields:

$$\sup_{\mathcal{P}_\delta} \mathsf{E}[Q(x,\boldsymbol{D})] = \min_{\lambda\geq 0}\left\{\lambda\delta + \mathsf{E}_{\mathsf{P_n}}\left[\sup_{\boldsymbol{D}}\{Q(x,\boldsymbol{D}) - \lambda|\boldsymbol{D} - \boldsymbol{D}_k|\}\right]\right\}, \tag{3.14}$$

Which is great news as we know how to estimate expected values in the empirical distribution, combining the original objective we get:

$$\min_{\lambda\geq 0}\left\{\tilde{c}x^\intercal + \lambda\delta + \frac{1}{n}\sum_{k=1}^{n}\sup_{\boldsymbol{D}}\{Q(x,\boldsymbol{D}) - \lambda|\boldsymbol{D} - \boldsymbol{d}_k|\}\right\}, \tag{3.15}$$

Recalling $Q(x,\boldsymbol{D}) = \min_{\boldsymbol{q}\geq 0} c\boldsymbol{q}^\intercal | A\boldsymbol{q}^\intercal + Bx^\intercal \geq (\boldsymbol{D}, \boldsymbol{0})^\intercal$, we can calculate the dual of this problem as $\max_{\boldsymbol{\eta}\geq 0} \boldsymbol{\eta}[(\boldsymbol{D},\boldsymbol{0})^\intercal - Bx^\intercal] \,|\, \boldsymbol{\eta}A \leq c$. Each of the terms $\sup_{\boldsymbol{D}}\{Q(x,\boldsymbol{D}) - \lambda|\boldsymbol{D} - \boldsymbol{d}_k|\}$ can be then written as $\sup_{\boldsymbol{D}}\{\boldsymbol{\eta}[(\boldsymbol{D},\boldsymbol{0})^\intercal - Bx^\intercal] - \lambda|\boldsymbol{D} - \boldsymbol{d}_k|\}$. Exchanging the sup and the max and isolating the terms depending on the vector $\boldsymbol{D}$ yields:

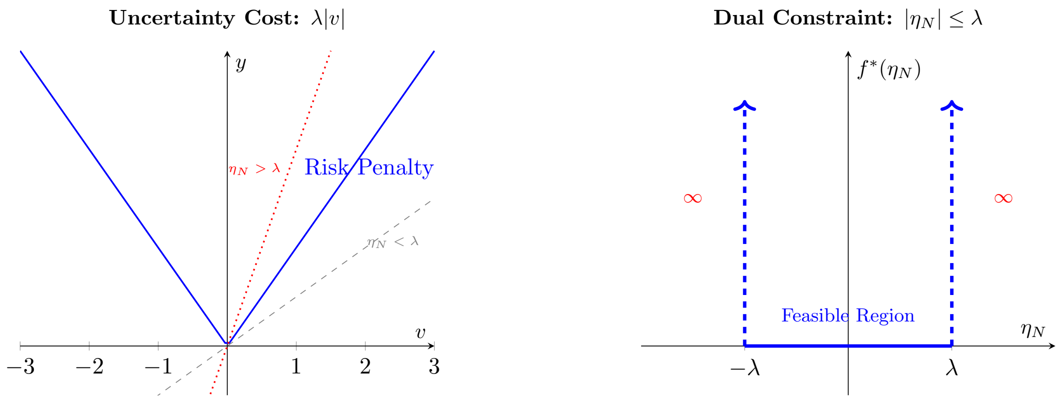
$$\max_{\boldsymbol{\eta}\geq 0, \boldsymbol{\eta}A\leq c} -\boldsymbol{\eta}Bx^\intercal + \sup_{\boldsymbol{D}}\{\boldsymbol{\eta}_N\boldsymbol{D}^\intercal - \lambda|\boldsymbol{D} - \boldsymbol{d}_k|\}, \tag{3.16}$$

Where $\boldsymbol{\eta}_N$ are the first N elements of $\boldsymbol{\eta}$ corresponding to the dual variables of the demand satisfaction constraints of the warehouse problem. Next, define the variable $\boldsymbol{v}_k = \boldsymbol{D} - \boldsymbol{d}_k$. The inner problem $\sup_{\boldsymbol{D}}\{\boldsymbol{\eta}_N \boldsymbol{D}^\intercal - \lambda|\boldsymbol{D} - \boldsymbol{d}_k|\}$ can be re-written as $\boldsymbol{\eta}_N \boldsymbol{d}_k^\intercal + \sup_{\boldsymbol{v}_k}\{\boldsymbol{\eta}_N \boldsymbol{v}_k^\intercal - \lambda|\boldsymbol{v}_k|\}$. The last term is the convex conjugate of the $L-1$ norm and is equal to 0 when $|\max \boldsymbol{\eta}_N| \leq \lambda$ or equal to ∞ otherwise. That it is equal to 0 is achieved by the constraints $-\mathbf{1}\lambda \leq \boldsymbol{\eta}_N \leq \mathbf{1}\lambda$. See Figure 3.6 for the intuition behind these constraints.

With this we can finally finish the rewrite of the problem as:

$$\begin{aligned}
\min_{\lambda,\boldsymbol{\eta},x} \quad & \left\{ \tilde{c}x^\intercal + \lambda\delta + \frac{1}{n}\sum_{k=1}^{n}(\boldsymbol{\eta}_N \boldsymbol{d}_k^\intercal - \boldsymbol{\eta}Bx^\intercal) \right\} \\
\text{s.t.} \quad & \boldsymbol{\eta}A \leq c, -\mathbf{1}\lambda \leq \boldsymbol{\eta}_N \leq \mathbf{1}\lambda, \\
& \lambda \geq 0, \boldsymbol{\eta} \geq 0, x \in \{0,1\}^M.
\end{aligned} \tag{3.17}$$

Which is almost a MIP except for the variables $\boldsymbol{\eta}Bx^\intercal$. The standard way of dealing with this, is to define new variables $z_{ij} = \eta_i x_j$ and add the constraints $z_{ij} \leq \tilde{M}x_j, z_{ij} \leq \eta_i, z_{ij} \geq \eta_i - \tilde{M}(1-x_j), z_{ij} \geq 0$ for a large enough $\tilde{M}$ (this is commonly known as the big-M method).



(a) Cost of deviation v

(b) Shadow Price "Discounting"

Figure 3.6: The relationship between deviation cost and shadow prices. In (a), if the shadow price η_N (the marginal cost of demand) is steeper than the risk penalty λ, the sup-problem becomes unbounded. In (b), this translates to a hard constraint: the robust optimizer only considers shadow prices "discounted" within the $[-\lambda, \lambda]$ interval, effectively capping the system's sensitivity to demand uncertainty.

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